



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

Give Me a Sign! Investigating the Brain Processing of Sign Language

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A Dissertation

Presented to the University of Dublin, Trinity College

in partial fulfilment of the requirements for the degree of

**Master of Science in Computer Science (Augmented & Virtual
Reality)**

Supervisor: Giovanni Di Liberto

August 2024

Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

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Give Me a Sign! Investigating the Brain Processing of Sign Language

Haozhe Ma, Master of Science in Computer Science
University of Dublin, Trinity College, 2024

Supervisor: Giovanni Di Liberto

This dissertation investigates how the brain processes sign language by examining EEG data from two groups: native French Sign Language (LSF) signers and French-speaking non-signers. The study hypothesises that LSF signers will show stronger neural responses to sign language stimuli compared to non-signers. Participants were shown videos in both normal and time-reversed playback conditions to observe differences in brain activity. Key findings reveal that LSF signers exhibit dynamic patterns of brain activity when viewing normal sign language videos, which reduced when videos are played in reverse. In contrast, the non-signers showed weaker neural responses, indicating limited comprehension of the visual language cues. Additionally, a new analytical metric, OpenPose-Instantaneous Visual Change (OpenPose-IVC), was explored but found to have some inconsistencies compared to traditional metrics. Future research will focus on improving this metric, expanding the participant pool, and further exploring the cognitive processes involved in sign language understanding. The results highlight the complex neural mechanisms behind sign language comprehension and provide a foundation for future studies to develop more advanced analytical tools using our novel, yet-to-be-published dataset.

Acknowledgments

Dissertation, right. I've always been interested in machines and how we think. Naturally, when I saw the project listing on the first day of this degree, I made my decision. Giovanni, my supervisor, you taught me way more than I could have imagined, from how to work as a researcher, to cultures in Europe, to career choices in the future. The weekly meetings were one of my favourite parts of this course, hearing PhD students sharing their new advancements and insights, which have been incredibly inspiring. My parents, ever since I set foot in this new country six years ago, you have always supported my stubborn and reckless choices. What I want to say to you is more than what gratitude can encompass. Whenever I'm happy, or having hard times, I know you are there to chat with, providing unwavering support and love. My friends from the course, you are not new friends to me anymore. You are the reason I'm still here; the memories from this one year feel like it's only the beginning of something lifelong. I had the deepest conversations and some of the most unintelligent ones with you, both of which I cherish forever. My friends around the world, you are the inspiration for this project. Whether it's a two-hour conversation about the brain in a hostel in Vienna, or learning the word 'footpath' in Istanbul, communication will stay as the most exciting part of my life, and I will keep exploring.

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University of Dublin, Trinity College
August 2024

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Chapter 1

Introduction

The perception of speech has been extensively studied due to its critical role in human communication and society, with a vast body of research dedicated to understanding the neural mechanisms underlying spoken language processing [19], [16]. However, despite its importance as a primary mode of communication for individuals who are deaf or have severe hearing impairments, sign language remains remarkably under-investigated in terms of its neural processing. This knowledge gap is particularly striking given that sign languages, such as French Sign Language (LSF), possess their own distinct grammatical structures, vocabularies, and syntax, which are separate from those of spoken languages [6]. Moreover, the visual-gestural nature of sign language presents a unique set of cognitive and neural processing demands, requiring the integration of visual, motor, and linguistic information. Consequently, the neural mechanisms involved in sign language comprehension are likely to differ significantly from those of spoken language. Understanding these mechanisms can offer valuable insights into the neural basis of human communication. For instance, it can help determine whether we pay the same amount of attention when understanding sign language compared to spoken language, and whether we use the same brain regions to process both manual and spoken languages. Furthermore, exploring the neural processing of sign language can illuminate the brain's capacity to support diverse modes of communication, highlighting its ability to facilitate the use of various languages and communication systems. By investigating the neural mechanisms of sign language comprehension, we can gain a deeper understanding of the complex interplay between language, cognition, and the brain, and contribute to the development of more effective communication strategies and interventions for individuals with hearing impairments. In collaboration with the Sign Language group at École Normale Supérieure in Paris, we are working with a unique dataset of neural signals recorded from individuals as they were presented with French Sign Language (LSF). The primary goal of this project is to

develop a framework for reliably identifying the neural signatures associated with sign language processing. To achieve this, we employ a combination of statistical methods and machine learning techniques to analyse the brain data. These advanced tools help us decode the neural representations of sign language and understand the cognitive processes involved. Additionally, we propose a novel metric that incorporates body pose estimation techniques. This approach provides an alternative way to analyse the visual-gestural components of sign language, allowing us to capture clearer movements. By potentially conducting separate analyses for different body parts, we aim to identify the specific neural mechanisms underlying sign language processing.

1.1 Structure of the Dissertation

In Chapter 1, we lay out the motivation behind the study, emphasising the under-explored area of neural processing in the context of sign language. This chapter also provides background on relevant topics such as speech comprehension, sign language, neural signals and neuroimaging techniques such as electroencephalography (EEG). This chapter also includes a review of closely related work, including the application of Multivariate Temporal Function (mTRF) Toolbox and studies investigating visual cortex entrainment to sign language.

In Chapter 2, the focus shifts to the design and implementation of this research. This chapter begins with formulating the research problem and our hypothesis. The design of the experiment covers aspects such as participant recruitment and stimuli design. The technical implementation is discussed with inclusion of feature extraction methods. The Chapter also addresses the pre-processing of EEG data and the overall analysis pipeline, which also considering ethical aspects and challenges encountered during the research process.

Chapter 3 presents the results and evaluations of the study. The analysis begins with an examination of a descriptive measure - GFP, and related statistical tests, followed by an exploration of Temporal Response Function (TRF) regression weights, topographical plots and prediction correlation analysis. The performance of a newly proposed metrics from this research is also reviewed in this chapter.

The dissertation concludes with Chapter 4, which provides a summary of our findings and their implications. Additionally, this chapter suggests future research directions, including the exploration of additional features from the novel dataset, building upon the results of this study.

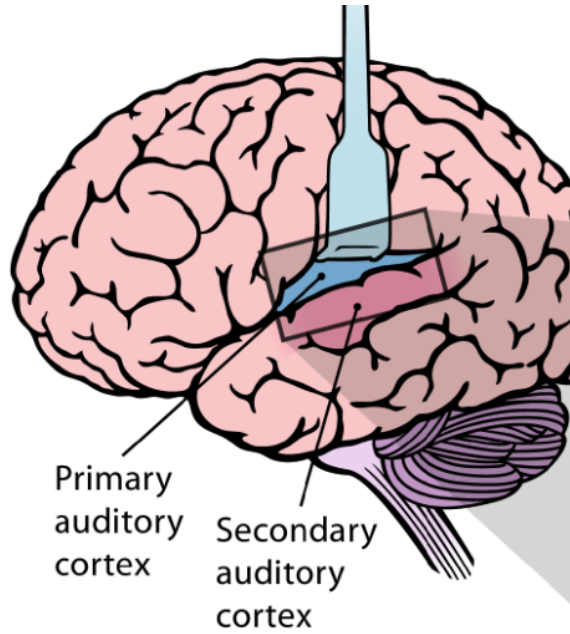


Figure 1.1: Illustration of the lateral view of the human brain with the auditory cortex exposed. Figure adapted from [8] Figure 1(B)

1.2 Background

1.2.1 Speech Comprehension

The ability to understand spoken language, a process fundamental for human communication and social interaction, involves complex cognitive process distributed across various brain regions. This comprehension relies on a hierarchical organisation of speech perception, with a succession of cortical areas processing progressively more abstract elements, from the sound of speech to meanings [16]. This hierarchical structure is supported by evidence, suggesting two main pathways for processing spoken language: a dorsal and a ventral pathway. The dorsal stream, consisting regions like the posterior superior temporal gyrus and the inferior parietal lobule, maps the auditory speech sounds to motor representations, playing a crucial role in speech production and phonological processing. Conversely, the ventral stream involves the middle and inferior temporal gyri, maps speech sounds to meaning to be understood, covers semantic and syntactic processing [19]. This distinction between dorsal and ventral pathways is relevant when considering potential similarities and differences in the processing of sign language, which may share commonalities in abstract processing while differing in sensory aspects.

The first step of auditory speech comprehension is the perception of input, it begins from the inner ear, where sound waves are converted to neural signals, these signals are then transmitted to the auditory cortex in the temporal lobe, they are on both sides of the

brain, above the ears. The primary auditory cortex processes the basic sound features, such as the pitch and volume, while higher-order areas in the superior temporal gyrus (STG) are involved in the more complex processing of the speech recognition [19]. As shown in Figure 1.1, where it illustrated the locations of primary and secondary auditory cortex. The primary auditory cortex is situated within the STG, and the secondary auditory cortex is adjacent and extends from the primary auditory cortex within the same gyrus.

Following the basic auditory processing, the brain engages in phonological processing, where phonemes - the neural representation of speech sounds is constructed by mapping acoustic signals to phonological representations. After this, the brain maps the phonemes onto stored lexical items (words) in the mental lexicon. This lexical access is happening in parallel with semantic processing, they retrieve and integrate the meaning of the spoken words [2].

To understand how words relate to each other within a sentence, the syntactic processing is engaged to allow the brain to interpret the grammatical relationships and hierarchical organisation of sentences. The Broca's area (the left inferior frontal gyrus) is involved in this process and works with the posterior superior temporal gyrus and other regions to complete syntactic parsing and interpretation [16].

The brain needs to integrate the semantic and syntactic information to form a coherent mental representation of a sentence it hears. The integration process allows the brain to advance from understanding the words and their meanings to also how these words combine to convey specific messages. In addition, contextual processing, or pragmatic processing will further refine the understanding of spoken language by incorporating the context, speakers' intentions and situational cues. This processing involves the right hemisphere and prefrontal cortex to interpret figurative language, idiomatic expressions and implied meanings [28].

In addition, previous studies proposed several theoretical models to explain the cognitive and neural mechanisms of speech comprehension. One influential model is the dual-stream model of speech language processing, which is generally explained above. It assume the existence of two main pathways for processing spoken language: the dorsal stream and the ventral stream. The dorsal stream involves regions such as the posterior superior temporal gyrus and the inferior parietal lobule, it is responsible for mapping the audio speech sounds to motor representations, facilitating the speech production and phonological processing part. The ventral stream, involving the middle and inferior temporal gyri, is responsible for mapping the speech sound to meaning, that includes semantic and syntactic processing [16, 19, 34, 35].

1.2.2 Sign Language

Sign language is a rich visual-manual language that is used by deaf communities around the world. Sign languages utilise hand shapes, movements, facial expressions and body postures to convey different meaning. Sign languages, like spoken languages, have their own grammar, syntax and lexicon, they are also not universal; different communities have developed distinct sign languages. The primary sign language used globally include American Sign Language (ASL), British Sign Language (BSL) and Chinese Sign Language (CSL) [30].

The sign language that we did the experiment on is the French Sign Language (Langue des Signes Française, LSF). It is the primary sign language used by the deaf community in France and partly in Switzerland. Its history trace back to the 18th century [4].

Sign language and spoken language exhibit differences in their modes of communication, structure. If we use LSF and English for comparison, LSF relies on hand shapes, movements, facial expressions and body postures to convey meaning, whereas English, an auditory-vocal language uses sounds produced by the vocal cords and understood through hearing. Structurally, LSF often uses a topic comment structure - a syntactic arrangement where the topic of the sentence is introduced first, followed by a comment or description of the topic. This language structure is different from the typical subject-verb-object (SVO) order found in English and French. For example, in English, the sentence “The cat is on the table.” have the subject “the cat”, followed by the verb “is”, and “on the table” is the object. In LSF, the same idea might be conveyed by first introduce the topic “the table” and then providing the comment (the cat is on it). This could be signed as “TABLE, CAT ON” in LSF where “TABLE” is the topic and “CAT ON” is the comment [15]. This structure in LSF allows its users to set the context or focus of the conversation before providing additional information.

Similarly, Austrian Sign Language (ÖGS) uses spatial indexing to establish the roles of arguments in a sentence. For example, Figure 1.2 shows a dynamic signed sentence in ÖGS, where the sentence “The girl helps the girl” is signed using spatial indexing to differentiate between the subject and the object. [27]

1.2.3 Neural Signal and EEG

Neural signals are the electrical impulses generated by neurons in the brain as they communicate with each other. These electrical signals are useful in understanding brain function and are typically measured in different neuroimaging techniques.

Neuralimaging techniques can be broadly categorised into non-invasive and invasive methods. Non-invasive techniques, such as electroencephalography (EEG), functional

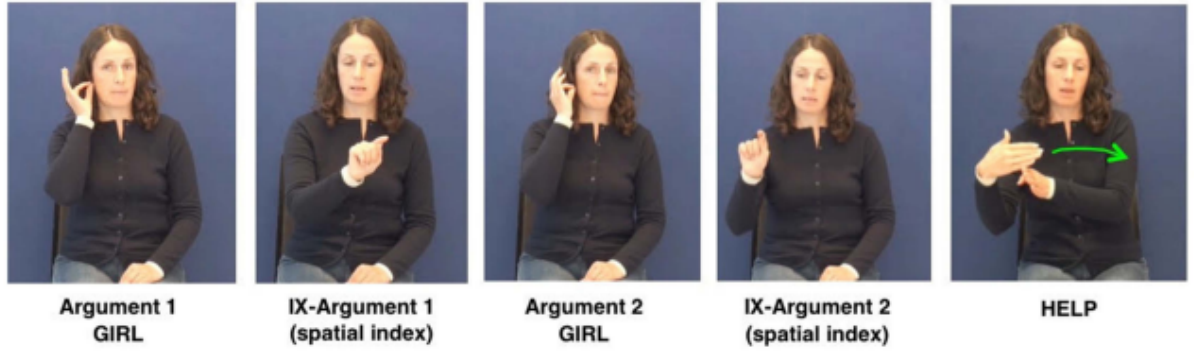


Figure 1.2: Dynamic Signed Sentence in ÖGS

Figure adapted from [27], The first frame shows the sign for ‘GIRL’ establishing the subject; the second frame uses a pointing gesture (spatial index) to establish the location of the subject in the signing space; The third frame ‘GIRL’ establishing the object; The fourth frame uses spatial indexing to establish location of the object; The fifth frame shows ‘HELP’ in sign, which is the action/verb in the sentence.

magnetic resonance imaging (fMRI) and magnetoencephalography (MEG), do not require a surgery or penetration of the skull, making them safer and more accessible options for brain function studies [9]. While these methods allow researchers to monitor brain activities with low risk of harming participants, in contrast, invasive techniques, such as electrocorticography (ECoG) and intracranial EEG (iEEG), involve the insertion and placement of electrodes on or within the brain tissue. While invasive methods mentioned provide higher spatial and temporal resolution and can record neural activity with higher precision, they are usually have limited accessibility due to their associated risks and ethical considerations [22].

One of the widely used methods and the method used in this project for measuring neural activity is electroencephalography (EEG). It is a non-invasive technique that measures the electrical activity in the brain through electrodes placed on the scalp. This methods captures the summed electrical activity of large populations of neurons, providing a high temporal resolutions of neural dynamics [24]. EEG’s high temporal resolution characteristic refers to its ability of the measuring technique to capture rapid changes in neural activity over time, this feature made it advantageous for studying cognitive processes since it can detect rapid changes in brain activity that occur in response to specific stimuli.

The EEG signal is often characterised by its frequency components, which are then categorised into different sub-bands: delta (0.5-4Hz), theta (4-8Hz), alpha (8-13Hz), beta (13-30Hz) and gamma (30-100Hz) [5]. Each of these sub-bands is associated with different types of brain activity and cognitive states. For instance, alpha waves are often linked to wakeful states, and beta waves are associated with active thinking and focus.



Figure 1.3: A mBrainTrain EEG cap placement on the author's head

In the context of this study, sign language comprehension, EEG can be used to investigate how the brain processes visual-manual linguistic information. Previous study have shown that specific EEG patterns are sensitive to linguistic processing in both spoken and sign languages.

BioSemi EEG Cap

In this study, our collaborators at PSL University in Paris employed the Biosemi EEG cap 64-channel system to capture the neural activity from participants. The Biosemi EEG cap, developed by the same named company based in the Netherlands, uses active electrode to enhance the quality of recorded EEG signals. Each electrode contains an amplifier that reduces noise and artefacts. When setting up the EEG cap, several preparation steps are involved to acquire optimal data from participants. In the experiment the author participated, which is a music related experiment from a PhD student from the same lab using similar setup, the EEG cap was carefully placed on the participant's head. Conductive gel is applied to each electrode to establish good electrical connection between the scalp and the electrode. The signal quality is then checked using Biosemi's software. Impedance level are monitored to ensure they are within acceptable ranges. During the recording session, participant is asked to stay still to avoid generating noise.

The system is capable of supporting sampling rate up to 16kHz to provide high temporal resolution, the recorded data is store in 24-bit BDF (Biosemi Data Format). The format consists of a header and data records. The header includes metadata like starting time, patient information and the number of data channels. Additional code are applied to perform segmentation and pre-process the BDF format for further analysis.

1.3 Closely-Related Work

1.3.1 The Multivariate Temporal Response Function (mTRF) Toolbox

Temporal Response Function (TRF) is a technique used to find relationship between time-locked stimuli and neural responses, it models the way the brain processes continuous streams of information by mapping stimulus features onto neural responses over time. The relationship between continuous stimuli such as speech or music and neural responses is often approximately linear over short timescales. EEG data have the characteristic of high temporal resolution, linear models can handle this multi points per second data efficiently without requiring extensive computational resources [10].

The toolbox supports both forward modelling and backward modelling, as illustrated in Figure 1.4, our project mainly utilises the encoding, which is the forward modelling part of the mTRF toolbox - we predict neural responses based on the stimulus, we assume that the neural response at a given time point is influenced by the stimuli at the current time and several preceding time points. This connection can expressed as a TRF, which describes how the stimulus is transformed into a neural response over time. The equation for forward model is given by:

$$r(t, n) = \sum_{\tau} w(\tau, n) s(t - \tau) + \epsilon(t, n) \quad (1.1)$$

where $r(t, n)$ represents the neural response at time t and channel n . $s(t - \tau)$ is the stimulus at time $t - \tau$. $w(\tau, n)$ is the TRF weight at time lag τ for channel n . $\epsilon(t, n)$ is the term of residual error at time t and channel n .

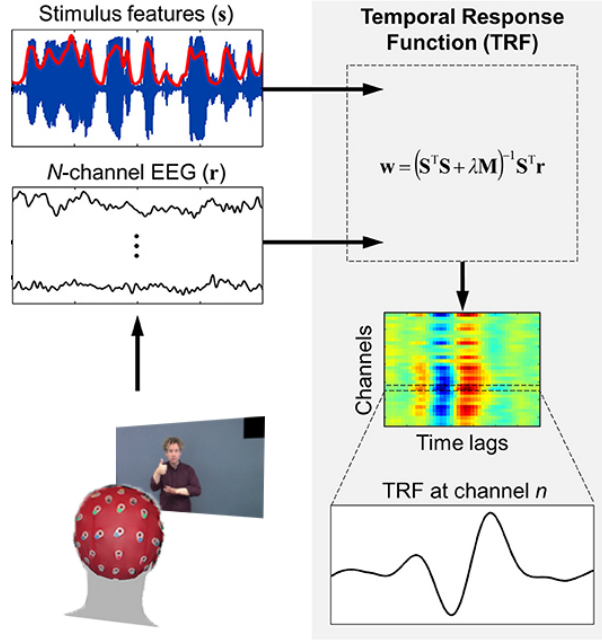


Figure 1.4: Schematic representation of the forward modelling approach implemented by the mTRF Toolbox.

Participants observe sign language video stimuli while their neural responses are recorded using multi-channel EEG(r). The stimulus features(s) are utilised to predict the neural response through the estimation of the Temporal Response Function (TRF). The TRF is computed using the equation. This forward modelling approach facilitates an examination of how visual stimulus features are encoded in the brain and aids in the interpretation of the underlying neural generators. Figure redrawn from [10]

To estimate the TRF weights $w(\tau, n)$, the toolbox uses ridge regression, it addressed the potential chance of over-fitting by including a regularisation measure, it has two terms, the first term represents the sum of squared errors (MSE) between the actual and predicted neural responses, and the second term is the regularisation term that penalises large weights to prevent the over-fitting. λ is regularisation parameter that balances the model fit and complexity.

The ridge regression solution for estimating TRF weights is given by:

$$\mathbf{w} = (\mathbf{S}^T \mathbf{S} + \lambda \mathbf{I})^{-1} \mathbf{S}^T \mathbf{R} \quad (1.2)$$

where $\mathbf{S}$ is the matrix of stimulus data, with each row representing a time-lagged version of the stimulus. $\mathbf{R}$ is the vector of neural response data. $\mathbf{I}$ is the identity matrix. The performance of the model is evaluated by comparing the predicted neural responses with the actual responses.

1.3.2 Visual Cortex Entrainment to Sign Language

In this project, two metrics are used for correlating with EEG signals. The first is adapted from Brookshire et al.'s paper [3]. In their paper, Brookshire et al. investigated the neural mechanisms underlying the processing of visual stimuli, specifically focusing on the phase-locking between EEG signals and visual changes in sign language. The neural entrainment concept mentioned in their paper refers to the synchronisation of neural oscillations with external rhythmic stimuli, such as visual or auditory inputs [31]. This process allows the brain to align its activity with relevant environmental cues, potentially facilitating sensory processing and attention selection. Their study employed coherence analysis to assess how consistently the brain's electrical activity aligns with the Instantaneous Visual Change (IVC) of sign language over a range of frequencies. The IVC metric represents the visual changes between video frames. It quantifies the variability in visual features in a way similar to audio envelopes.

The findings suggested significant coherence between EEG signals from the participants that watched sign language stimuli videos and IVC in the 0.4 to 5 Hz range, with a peak at 1 Hz. This indicates that the brain entrains to the rhythmic patterns of visual stimuli. They further concluded that the brain's ability to entrain to low-frequency signals is a crucial aspect of sensory processing - this entrainment was observed not only in response to visual stimuli but also consistent with findings in previous auditory research, where the brain shows similar phase-locking to the rhythm and pitch of sounds [1][25][18][32]. The implication of these studies suggests that low-frequency entrainment might be a mechanism that the brain uses to process and integrate sensory information across different modalities. As Obleser and Kayser point out, neural entrainment plays a crucial role in attentional selection, helping the brain to focus on relevant sensory inputs while suppressing irrelevant ones [31]. The coherence at lower frequencies facilitates the synchronisation of neural oscillations with the outside environment (e.g., speech communication, sign language reception), thereby enhancing perceptual and cognitive functions [3].

This study is closely relevant to our project as it provides a groundwork on studying how the brain synchronises with visual changes. Part of our study aims to explore the temporal dynamics of EEG responses to IVC using the multivariate temporal response function (mTRF) approach, explained in Chapter 1.3.1. By shifting the focus from phase-locking to temporal correlation, we seek to offer a complementary perspective on the processing of visual stimuli, potentially offering new insights into the temporal characteristics of brain responses to visual stimuli.

Chapter 2

Design & Implementation

This chapter outlines the design and implementation of our study on neural responses to sign language stimuli. We present our hypothesis, experimental design involving two cohorts (LSF signers and French speakers), and detailed participant recruitment strategies. The chapter covers stimuli design, including the creation of sign language videos with synchronisation markers.

Key implementation aspects are also discussed, including the extraction of Instantaneous Visual Change (IVC) and the novel OpenPose-IVC feature. We detail the EEG data pre-processing pipeline and the conversion to the Continuous Neural Data (CND) format. Ethical considerations of the data used in this project are addressed, including approval from the relevant ethics committees in Paris and submission to the Trinity SCSS ethics committee.

2.1 Hypothesis

Some previous works have investigated the brain’s entrainment to the visual signals of sign language. It has been shown that brain activity synchronises with spoken language’s rhythmic patterns and various linguistic properties of the stimulus [14, 33]. Previous work also shows that spoken and sign language rely on the same neurological systems [15, 26]. However, few studies have investigated neuronal responses to continuous, natural signals in sign language [3].

Previous studies provided insights into behavioural results and the frequency-specific phase relationships between EEG and visual stimuli. Our study builds on these existing findings and focuses on the correlation between EEG and stimuli in the temporal domain. In this first pilot study using the new dataset, we aim to establish that we can measure the neural signatures of how the brain processes sign language and identify the

best-correlated features that can be used to further test other hypotheses on sign language processing [10].

To test this, we need to design an experiment that can compare and quantify the neural responses to continuous sign language stimuli. We will need two cohorts of subjects: one group that understands LSF sign language and another that does not. We will measure and compare the EEG responses between the two groups while they observe continuous sign language videos.

An additional control is needed to test the specificity of the neural responses to meaningful sign language stimuli. For both cohorts, we will include normal playback and time-reversed playback of our sign language video stimuli. This control can help eliminate the possibility that the observed neural responses correlations are merely due to low-level visual features or general motion patterns rather than the linguistic content of the sign language.

By comparing the EEG responses to normal and time-reversed playback, we can test whether the brain’s entrainment is specifically tuned to the temporal structure and linguistic properties of the sign language. If the neural signal correlation levels are significantly different between normal and time-reversed playback, it would suggest that the brain is processing the temporal and linguistic information in the sign language, rather than just responding to visual motion. We will analyse the correlation between EEG signal and the sign language stimuli features across different EEG sub-bands, such as 1-30Hz, 1-8Hz. By examine the results from these sub-bands, we aim to identify which frequency ranges are most strongly associated with the neural processing of sign language.

As illustrated in Figure 2.1, **our hypothesis is that the neural response in the signer’s group will be the highest and most correlated to the features extracted from the stimuli videos when watching the correctly played back videos. We assume that this group, due to their familiarity with sign language, will exhibit stronger and more specific neural entrainment to the temporal and linguistic properties of the sign language stimuli compared to the non-signers’ group.**

2.2 Experiment Design

In the experiment, participants will watch videos in French Sign Language (Langue des Signes Française, LSF) on a computer monitor with no auditory input. Since the study’s aim is to study the cortical tracking of linguistic properties in sign language and compare it with cortical tracking of spoken language, the participants’ brain responses are recorded using an Biosemi EEG (electroencephalogram) cap with 64 electrodes. A small amount of electrolyte gel is applied to the scalp for each electrode to facilitate signal transmission.

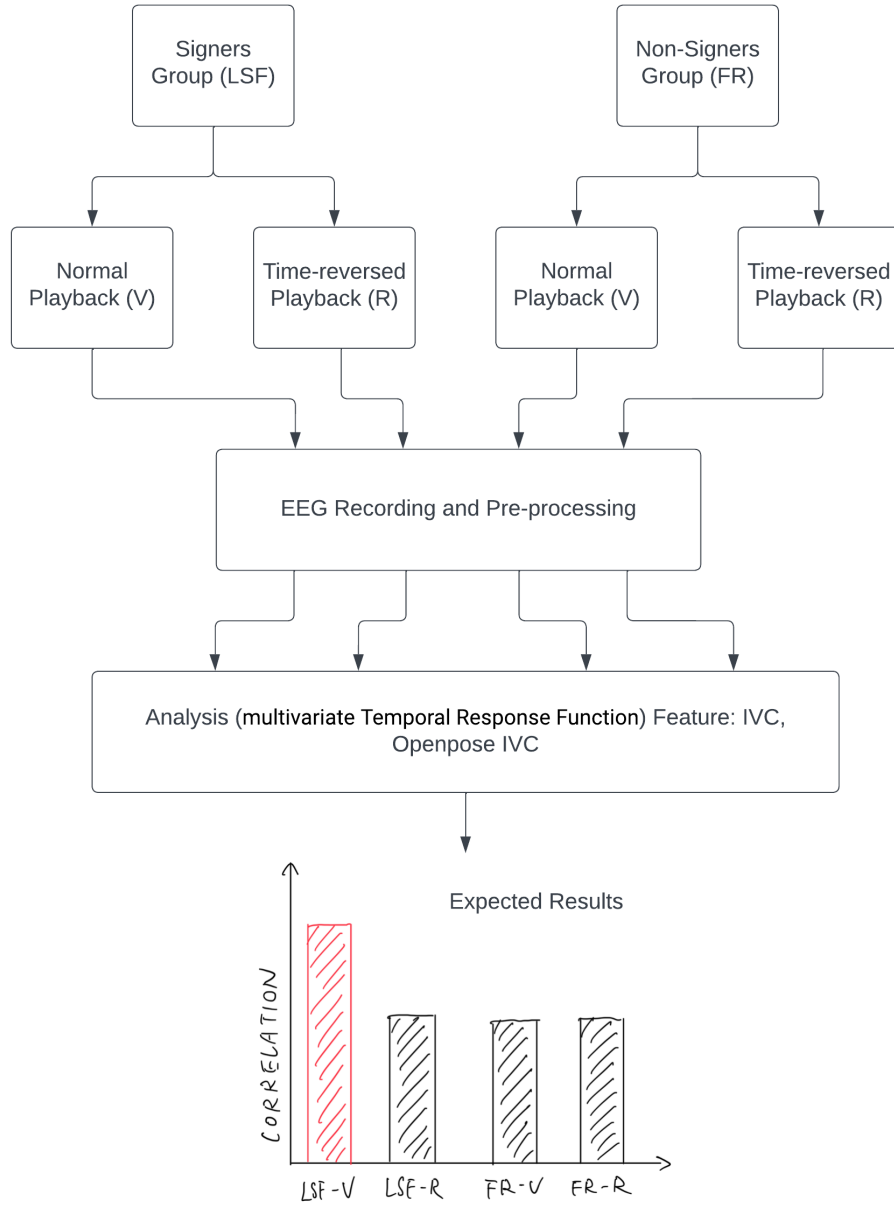


Figure 2.1: Experimental Design and Expected Results for EEG Study on Sign Language Processing

This figure illustrates the experimental design for investigating the neural responses to sign language stimuli. The study involves two cohorts: the Signers Group (LSF) and the Non-Signers Group (FR). Each group is exposed to two conditions: Normal Playback (V) and Time-Reversed Playback (R) of sign language videos. EEG signals are recorded and pre-processed for both groups under these conditions. The analysis focuses on the correlation between EEG signals and video features, specifically using the mTRF method with Instantaneous Visual Change (IVC) and Openpose IVC features. The bottom graph represents the hypothesis, showing that the Signers Group (LSF) is expected to exhibit the highest correlation with the normal playback stimuli (LSF-V), while lower correlations are anticipated for the other conditions and the Non-Signers Group (FR).

The gel will be washed off with water at the end of the experiment. Each participant are placed in a soundproof booth to minimise external noise and distractions. At the end of each video, participants will answer a short list of questions to qualitatively measure their attention levels.

2.2.1 Recruitment of Participants

The study aims to recruit a total of 50 participants, evenly divided between two groups: fluent signers of Langue des Signes Française (LSF) and native French speakers (FR). All participants are required to be between 18 and 50 years old and to have good or corrected vision (e.g., glasses or contact lenses).

As of the time of writing, data collection is ongoing. To date, we have recorded data from 22 participants, comprising 6 fluent signers of LSF and 16 native French speakers. Recruitment efforts continue to ensure a balanced representation of both groups in the final dataset.

Deaf Participants

For the LSF signers, selection is based on their fluency in French Sign Language. If participants use a cochlear implant or hearing aid, these devices will be removed during the experiment. Participants are recruited through various channels to ensure a diverse and representative sample. These channels include social networks (e.g., LSF groups on Facebook), personal networks (e.g., friends, Deaf consultants), and professional networks (e.g., former participants who have consented to be contacted for further experiments). An invitation video in LSF is sent via email to the selected Deaf participants, along with a list of contact options. During the online call for scheduling, participants are provided with information such as signed names, acknowledgements of their participation, and the criteria for participation. Once the date and time are confirmed, a confirmation email is sent, including an instructional video and detailed information about the experiment.

Hearing Participants

The hearing participants are recruited via the RISC (le relais d'information sur les sciences de la cognition), an online platform used for recruiting participants for cognitive science experiments with personalized criteria. They are contacted by email and provided with a questionnaire in written French to ensure they are French monolinguals. To ensure the ethical standards and compensate for participants' time and efforts, the collaborators at PSL University are paying every participants €15 per hour, regardless of whether participants complete the entire session or choose to withdraw early. The provision of



Figure 2.2: Frame capture from one of the stimuli video V01

compensation is intended to respect participants' time and ensure voluntary participation without any undue pressure.

To protect participants' privacy and confidentiality, an anonymisation protocol is employed. Upon registration, each participant is assigned with a unique identifier (ID). A secure, private document, accessible exclusively to the principle investigators at PSL University, contains participants' details. All sociological, behavioural and physiological data collected during the study are recorded using only this unique ID, ensuring participant anonymity throughout the research process. This identifier is never disclosed to external entities including researchers at Trinity, thereby maintaining confidentiality.

2.2.2 Stimuli Design

The sign language stimuli videos are purchased with authorisation of use with scientific purposes. The experiment comprises 14 original stimuli videos, each varying in duration from 2 to 8 minutes.

As shown in Figure 2.2, a frame capture from one of the videos, in all the stimuli videos, the sign language performer stand against a solid blue background wearing a burgundy shirt. The framing of the videos are from about the waist up, focusing on the upper body, ensuring all gestures and facial expressions are clearly visible. The lighting is even, without much shadows casting on the performer's face when doing hand movements.

One feature of these videos is the black rectangle located in the top right corner of the screen, serving as a synchronisation marker for EEG data collection. This marker allows for precise alignment of the video stimuli with the EEG recordings, ensuring that neural responses can be mapped correctly to specific movements in the video. Each video includes

a one-second pause at the beginning, where the the video freeze at first frame and the rectangle on top right corner is white, it is a deliberate design choice to facilitate the synchronisation, it helps in avoiding any initial artefacts that might occur when the video starts playing. Despite the pause, the trigger for the EEG data collection is set to time zero, meaning it coincides exactly with the start of the video, ensuring that there is no delay in the synchronisation.

The experiment includes a control group of time-reversed stimuli, essentially reversing all the original videos. All the videos that presented to the participants are encoded and compressed using HandBrake. The original videos are recorded at 60 frames per second but are encoded at 30 frames per second for the actual experiments. The video files are in a resolution of 1280x720 pixels and encoded using the H.264 codex.

For each participant, 14 trials are conducted, with the order of the trials randomised. Each participant watches 11 trials in the correct playback condition and 3 trials in the time-reversed condition. The order of these trials is recorded in a CSV file per each participant.

2.3 Implementation

2.3.1 Feature Extraction: Instantaneous Visual Change (IVC)

The Instantaneous Visual Change (IVC) is a novel metric developed by Brookshire et al. [3] to quantify visual changes over time, specifically within the context of sign language. This metric was introduced to facilitate the examination of neural entrainment to visual rhythms in sign language, analogous to how the acoustic envelope is used to study auditory rhythms in speech, see Figure 2.3.

This feature extraction process starts with frame differencing, where the differences between adjacent grey-scale video frames are calculated for each pixel. The changes in brightness between frames are detected, next, the pixel-wise differences are squared, aggregating both increases and decreases in brightness and resulting all changes are considered positively in the summary metric. The squared differences are then summed across all pixels for each pair of adjacent frames. This summation results in a single value representing the magnitude of visual change between two frames. Finally, these values forms a time series that reflects the amplitude of visual changes over time.

The mathematical formula of IVC is given by:

$$IVC(t) = \sum_i [x_i(t) - x_i(t - 1)]^2 \quad (2.1)$$

In formula 2.1, $x_i(t)$ is the gray-scale value of pixel i at time t . The implemented function calculates the visual change from one frame to the next, providing a metric for the amount of motion or change in the visual scene over time. This feature draws a parallel to the acoustic envelope of spoken language, which is a highly reduced representation of sound that traces extreme amplitude values in the time-varying signal. The acoustic envelope depends on the movements of various vocal track components, such as the rhythms opening and closing of the mandible.

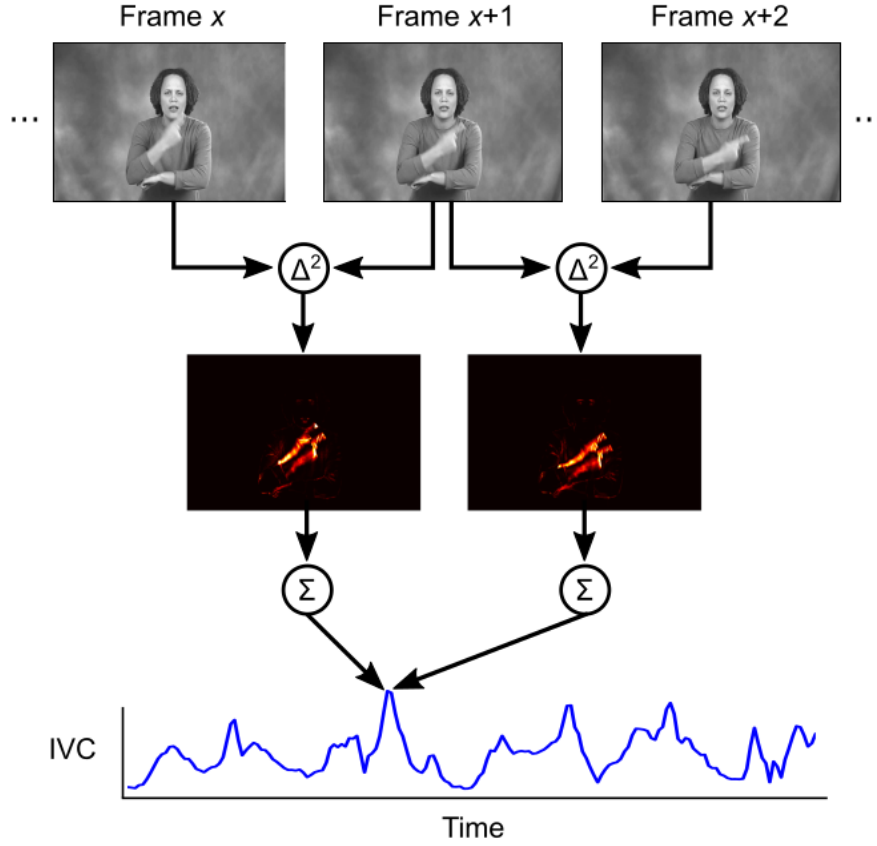


Figure 2.3: Calculation of the Instantaneous Visual Change (IVC)

This figure illustrates the process of calculating the IVC, which summarizes the total visual change at each point in time. First, the difference between adjacent grey-scale video frames (Top) is calculated for each pixel. To aggregate both increases and decreases in brightness, these pixel-wise differences are squared (Middle). Finally, the brightness values in all pixels of the squared-difference images are summed to obtain a single value summarising the magnitude of change between two video frames. Computation of this value for each adjacent pair of frames yields a time-series (Bottom).

Figure adapted from Brookshire et al. [3]

Temporal Structure Characterisation and Comparison to Spoken Language

Brookshire et al.'s analysis of the IVC reveals quasi-periodic oscillations, showing that rhythmic visual changes driven primarily by the movements of the signer's hands and arms, as well as the face, head and torso movements. They computed the power spectra of the IVC to characterise variability within and across different sign language - four sign languages were used: American Sign Language (ASL), German Sign Language (DGS), British Sign Language (BSL) and Australian Sign Language (Auslan). The analysis shows that the power decreases monotonically with increasing frequency, without pronounced peaks.

When compared with the broadband envelope of spoken language, the IVC shows significant difference in temporal structures. Spoken language has stronger modulations at frequencies greater than 2 Hz, due to the rhythmic nature of syllables, which occurs at approximately 2-10 Hz. In contrast, sign language displays more pronounced modulations at lower frequencies, reflecting the slower movements of sign articulators (hands) compared to that used in speech (vocal tract).

Brookshire et al. used this feature and EEG to investigate the neural entrainment to visual rhythms in ASL speaker when they watch ASL videos, their results indicate **significant cortical coherence to visual oscillations in the IVC, particularly in delta band (1-4 Hz) and theta band (4-8 Hz), with a peak around 1 Hz, strongest over the occipital and parietal cortex. Non-signers also exhibits coherence to sign language with less pronounced at frontal sites compared to fluent signers. These findings suggest that there is significant involvement of visual processing regions in fluent signers' brains when they watch ASL videos, supporting their hypothesis that low-frequency oscillatory entrainment serves as a general cortical mechanism for enhancing perceptual sensitivity to rhythmic information across sensory modalities.**

Implementation of IVC Using Python in This Project

In our project, we implemented the IVC algorithm in Python, it reads all the video files in given folder, does the frame difference and summation calculation and use `numpy` and `scipy.io` to save the time series as MATLAB and NumPy file format respectively. In our project's analysis pipeline, we only used the MATLAB format. If a stimuli video has 8,000 frames, the IVC feature file will be in the shape of 1x8,000 double in MATLAB. As the stimuli videos that we used are 30 frames per second (30 Hz), and the decision was made to down-frequency the EEG recording to 64 Hz, the IVC feature files need to be resampled to 64 Hz to match with the recording. In `np2resampledmat.py`, we

performed the resampling of IVC feature. The script iterates over the `.npy` files in the IVC output directory, load each file using the `resample` and `resample_poly` functions from the `scipy.signal` module. We tried both resampling methods, `resample` function performs Fourier method resampling and the `resample_poly` performs uses poly-phase filtering for resampling.

Since there is a one-second pause at the beginning of each correct playback stimuli video, an additional (optional) padding is added to the start of the resampled data, if the file name starts with 'R', suggesting the IVC file is from reversed video. It will add 63 zero values to the front and removes the 63 samples. Finally, the resampled data is saved as `.mat` files in the output directory.

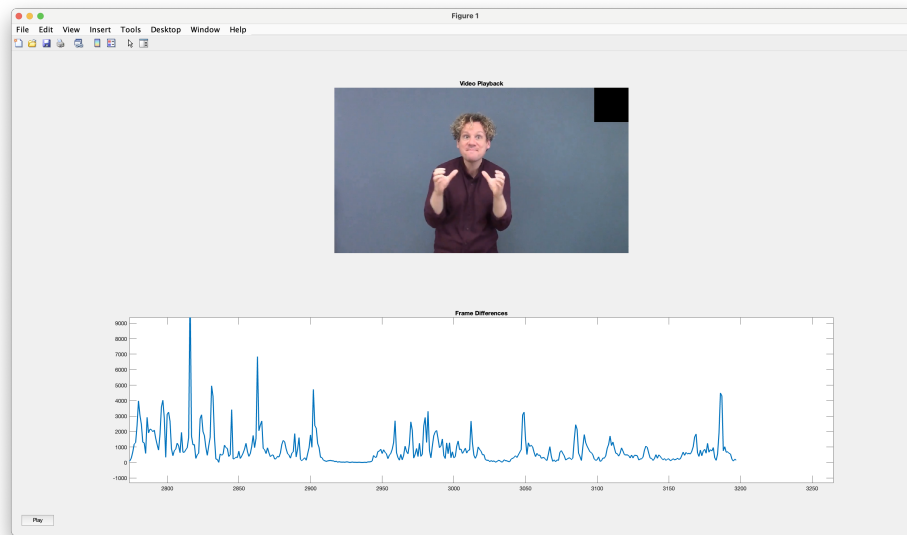


Figure 2.4: IVC signal playing against stimuli video.

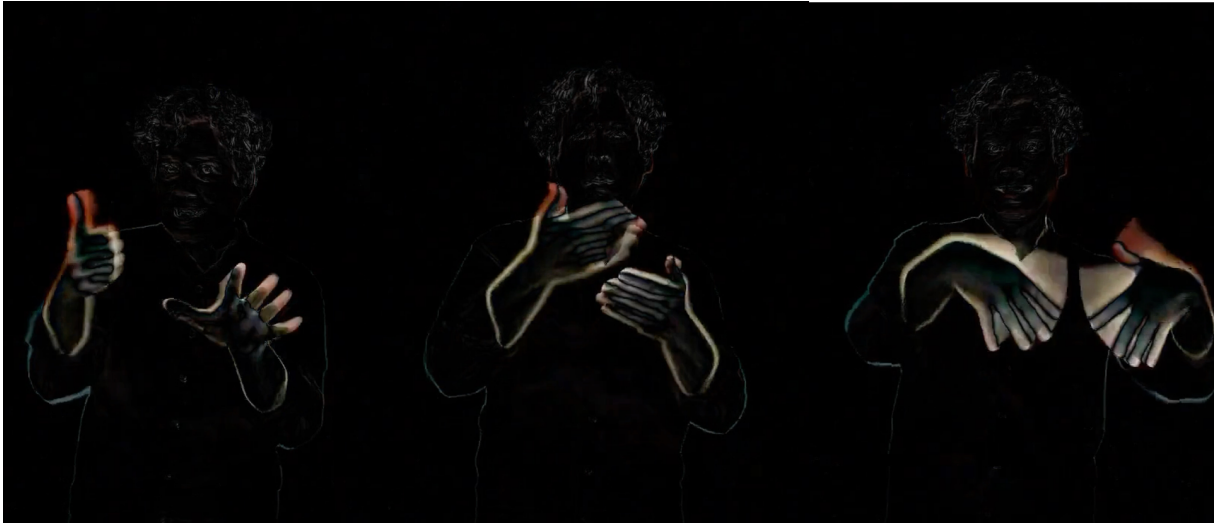


Figure 2.5: An illustration of IVC on our stimuli video

Demonstration of the frame difference was achieved with Adobe Premiere Pro: duplication of video track, use 'difference' blending mode on the top video track and move the two video tracks off by 1 frame.



Figure 2.6: OpenPose Render of Body, Hand and Face 2D Pose Estimation

2.3.2 OpenPose-IVC

We propose a new OpenPose-Incorporated Visual Change (OpenPose-IVC) metric by integrating pose estimation with the Instantaneous Visual Change (IVC) methodology. This new metric leverages the OpenPose framework, which was introduced by Cao et al. [7], designed for real-time detection of human body poses using a network that employs Part Affinity Fields (PAFs) and a multi-stage architecture. The primary strength of OpenPose lies in its ability to provide detailed spatial information about body movements, including key points corresponding to arms, hands, torso, and facial expressions, with optional capability of 3D pose estimation.

The OpenPose model operates on a bottom-up approach, allowing it to simultaneously detect multiple individuals within a single frame without a predetermined limit on the number of people, a capability not typically afforded by traditional top-down methods. The PAFs facilitate the association of detected body parts with individual identities, ensuring high accuracy in pose estimation even amidst overlapping figures. This method also allows for efficient computation, enabling the calculation of this feature quickly on standard computer hardware.

IVC as a way of extraction envelope like feature from sign language video by quantifies the difference between consecutive frames provides a viable approach to create feature that is similar to speech study. The approach, however, does not take into account the spatial

relationships between different body parts and does not capture the nuanced movements that inherent to sign language communication. The movements of joints and body parts in sign language are directly related to the semantic content being conveyed, making them more relevant to cognitive processing[29]. By focusing on these movements, we are more likely to capture the meaningful aspects of the sign language that the brain is processing. Additionally, pose estimation can help isolate linguistic features such as hand shapes, movements, and facial expressions, which are crucial for understanding sign language, as illustrated in Figure 2.6.

Our study also aims to go beyond replicating the findings of Brookshire et al. [3]. Our final dataset will include word-by-word transcription of the sign language video, which will provide a unique opportunity to integrate linguistic features into our analysis. To fully leverage this data, we need a metric that can capture the linguistic characteristics of sign language and combine it with our new transcription feature. The OpenPose-IVC metric has the potential to provide a more nuanced understanding of the relationship between sign language and EEG signals, and can be used to investigate questions such as: How do linguistic features such as hand shape, movement, and facial expressions influence the neural processing of sign language? By developing a more sophisticated metric for analysing sign language, we can gain a deeper understanding of the complex interplay between visual, linguistic, and cognitive processes involved in sign language comprehension.

Pose Key Point Extraction

For each frame in the sign language video, OpenPose outputs a structured JSON format that includes an array of body key points. Each key point is defined by its (x, y) coordinates and a confidence score indicating the reliability of the detection. A sample JSON output for a single frame is illustrated below:

```

{
  "version": 1.3,
  "people": [
    {
      "person_id": [-1],
      "pose_keypoints_2d": [
        627.44, 227.754, 0.893681,
        617.625, 384.449, 0.81887,
      ],
      "face_keypoints_2d": [],
      "hand_left_keypoints_2d": [],
      "hand_right_keypoints_2d": [],
      "pose_keypoints_3d": [],
      "face_keypoints_3d": [],
      "hand_left_keypoints_3d": [],
      "hand_right_keypoints_3d": []
    }
  ]
}

```

The OpenPose output includes various categories of key-points: body key-points, facial key-points, and hand key-points, allowing for a comprehensive analysis of the signer’s physical expressions and movements. This granularity in data is particularly beneficial for examining subtle nuances in sign language, as specific facial expressions can alter the meaning of signed gestures.

Quantification of Motion Through OpenPose-IVC

To calculate the OpenPose-IVC, we take a novel approach that involves calculating the variance in key point positions across consecutive frames. For each key point indexed as K_j (where j indexes the key points), the Euclidean distance between the positions in frame t and frame $t - 1$ can be computed using the formula:

$$d = \sqrt{(x_{j,t} - x_{j,t-1})^2 + (y_{j,t} - y_{j,t-1})^2} \quad (2.2)$$

where $(x_{j,t}, y_{j,t})$ and $(x_{j,t-1}, y_{j,t-1})$ represent the coordinates of key point j in consecutive frames.

After computing the distances for each key point, these values are squared and aggregated

to yield a single OpenPose-IVC value for each frame transition:

$$OpenPose - IVC(t) = \sum_j d_j^2 \quad (2.3)$$

This total variance value indicates the magnitude of movement among the relevant key points, thereby providing an aggregate measure of visual change specifically associated with sign language articulation.

Normalisation and Confidence Scoring

To enhance the quality and reliability of our OpenPose-IVC calculations, we incorporated normalisation strategies that account for various video characteristics, such as frame rate and resolution, as well as the inherent variability in key point detection reliability. One aspect of our normalisation process involves using the confidence scores provided by OpenPose’s output. These scores help us weight the contributions of corresponding key points, thereby minimising the influence of outliers. Specifically, we apply a threshold (generally set around 0.6) to filter out key points with low confidence scores, which are deemed unreliable and could introduce noise into the motion metrics. By doing so, we ensure that only key points with higher confidence scores contribute significantly to the OpenPose-IVC calculations, leading to more accurate and robust measurements of motion. The final result is a time series data, which is also saved in both MATLAB and NumPy formats for subsequent analysis.

2.3.3 EEG Data Pre-processing

In Chapter 1.2.3 and 2.2, we explained the experimental setup for the project and the BDF file format for storing the EEG recording data.

We have a Python script `segment_and_preprocess_EEG.py`, it outlines a comprehensive pre-processing pipeline to prepare the raw EEG data from recording sessions for later analysis.

File Handling and Subject Selection

The script imports common data manipulation libraries such as NumPy and Pandas, along with MNE for EEG data handling. The script load the project directories and searches for BioSemi’s ‘.bdf’ files within the EEG directory.

Raw EEG Data Directory Structure

```
EEG_raw/EEG
  127740.bdf
  127740_signlanguage_2023-08-31_14h03.11.325
    127740_signlanguage_2023-08-31_14h03.11.325.csv
    127740_signlanguage_2023-08-31_14h03.11.325.psydat
    answers_to_questions.csv
    log.log
    stimuli_sequence.csv
  ...
```

Under the root directory ‘`EEG_raw/EEG`’, serves as the repository for all raw EEG recordings and associated metadata. Each subject’s data, as mentioned in Chapter 2.2.1, are fully anonymised and has an individual folder named after their unique identifier, which is a six-digit number, such as ‘164123’, ‘910810’.

Within each subject-specific sub-directory, the raw EEG data is stored in a BioSemi Data Format (BDF) file, named using the subject’s identifier followed by ‘`.bdf`’ extension. These files contains the continuous EEG recordings captured during the experiment sessions. On the same level, there is a sub-folder named with the subject’s identifier and a timestamp indicating the date and time of the recording. These sub-folders contains several files that provides additional context and metadata for our pre-processing.

The ‘`stimuli_sequence.csv`’ within each sub-folder details the sequence of stimuli presented to the subject during the recording sessions. As mentioned in Stimuli Design part (2.2.2), there are 14 stimuli videos in both correct playback (V) and time-reversed playback (R) in random order. This file is important for aligning the EEG data with the experimental conditions. The ‘`answers_to_questions.csv`’ file captures the subject’s responses to behavioural questions posed during the sessions. Additionally, the folder contains a ‘`log.log`’ file that records all the session’s events, for example, key press and program outputs.

The script iterates over these files, extracting the subject ID from each file name and only selects files corresponding to the subjects listed in ‘`subjects_to_process`’ are processed - this is how we separate the two cohorts (FR and LSF).

Data Segmentation and Pre-processing

The raw EEG data is loaded using MNE's `'read_raw_bdf'` function. The data is loaded into memory and the channel montage is set to the standard 10-20 system. The 10-20 system provides a standardised method for electrode placement, it refers to the distance between adjacent electrodes are either 10% or 20% of the total front-back or right-left distance of the skull. Events in the trigger channel are identified by using MNE library's `'find_events'` function. Our script only select relevant events corresponding to the start of the stimuli presentations. (Synchronisation of the video stimuli refer to Stimuli Design part 2.2.2) The onset times of these events are calculated and compared with number of sequence for verifying purpose.

For each stimulus (sign language video), the script crops the data to the duration of the stimulus presentation. Then applies several pre-processing steps. It start by doing channel selection, excluding the presumed empty, non-EEG channels to focus on the 64 EEG channels. Then applied notch filtering to remove the noise at 50 Hz, this step can eliminate interference from electrical sources. A band-pass filter is applied to isolate the frequency range of interest. As we are investigating more than one sub-bands of EEG, this can be set to sub-bands we want to prepare, such as delta band (1-4 Hz) and theta band (4-8 Hz). The script then applies interpolation for bad channels to mitigate the impact of noisy of faulty electrodes.

The data is referenced in the next step, using either the average of the mastoid electrodes or the average of all channels, in our case, we only select 'Mastoids' in our experiments. Re-referencing is a step in EEG data processing that involves recalculating the voltage at each electrode relative to a new reference point. The reference points are typically chosen to minimise the noise and enhance the interpret-ability of the EEG signal. Mastoid Re-reference, uses the average of the signals from the mastoid electrodes (which are typically placed behind the ears) as reference, they are often chosen because they are relatively inactive compared to the brain regions of interest, provide a reference point.

The last step of the pre-processing is down-sampling, the recording data is down-sampled to 64 Hz to reduce the computational load and storage, also making it easier to match with our video stimuli with relatively low sampling rate.

Pre-processed Data Storage

The pre-processed data and the steps applied are stored in MATLAB format. Under the `'EEG/subband/Mastoids'` folder, all the subjects' in that specific band-passed filtered range are stored in the same level. Each subject has an individual folder named according to their identifier ID. These folders contains data files that stored in `' .mat '` format, with

filenames following a convention where ‘R’ denotes the time-reversed stimuli tracks and ‘V’ are the correct playback tracks (e.g. ‘R08.mat’, ‘V14.mat’).

In addition to the ‘.mat’ files, each subject folder includes three CSV files:

‘answers_to_questions.csv’, ‘preprocessing_pipeline.csv’ and ‘stimuli_sequence.csv’. The pre-processing pipeline file contains the order and all meta information of the pre-processing steps that explained above, the ‘answers_to_questions.csv’ and ‘stimuli_sequence.csv’ file contains the sequence of the stimuli presented to the subject during the recording.

2.3.4 Continuous-event Neural Data (CND) and Conversion

The Continuous Neural Data (CND) structure is a standardised data framework developed by the Cognition and Natural Sensory Processing Initiative (CNSP) [23] to facilitate the pipeline of neural data acquisition during continuous sensory experiments. These experiments usually involves uninterrupted streams of stimuli over extended time period. the CND structure is designed to solve the fragmentation and lack of standardisation in existing data formats [13].

Organisational Framework

The CND structure prescribes a organisation for datasets, both for folder hierarchy and data structure. The recommended folder structure for typical CND dataset include the root directory named after the dataset (e.g. SignLanguageProject), in which contains three main folders: **eeg**, **stim** and **dataCND**. The **eeg** folder contains raw neural data files (e.g. ‘.bdf’) that need to be pre-processed, while the **stim** folder contains the stimulus data files (e.g. ‘.wav’ files for speech, music stimuli).

The **dataCND** folder contains pre-processed and synchronised neural and stimulus data files in MATLAB format (‘.mat’), see Chapter 2.3.3. The key files within the **dataCND** folder are structured as follows.

The **dataStim.mat** file contains the pre-processed stimulus features, in our project, it would be the IVC and Openpose-IVC features. The features contains name, data, fs (frequency), stimIdxs, condIdxs and condNames (condition names). The names field is a cell array of character arrays labelling each feature set (e.g., ‘speech envelope’, ‘word onsets’), while the data field is a cell array where each cell contains a matrix representing the stimulus feature vectors for each trial, with dimensions corresponding to time samples and feature dimensions. The ‘fs’ field represents the sampling frequency of the stimulus data, and the ‘stimIdxs’ and ‘condIdxs’ fields indicate the indexes of the stimuli and conditions respectively. The ‘condNames’ field contains the names of the conditions (e.g., ‘V’, ‘R’).

The neural signal recorded during the stimulus presentation are stored in separate files, the naming conventions as `dataSub1.mat`, `dataSub2.mat`, ..., `dataSubN.mat`. Each of the file includes the neural structure with the following fields: `dataType`, `deviceName`, `fs`, `data`, `origTrialPosition`, `chanlocs`, `extChan` and `cndVersion`.

The `dataType` field labels the type of response measure, in our project, it is ‘EEG’ for electroencephalogram data, while the `deviceName` represents the name of the recording system (e.g., ‘BioSemi’). `fs` field is the sampling frequency of the EEG data, which was down-sampled to 64 Hz, and the `data` field contains a cell array where each cell contains a matrix representing the neural data for each trial, with dimensions corresponding to time samples and channels. The `chanlocs` field contains the channel locations in EEGLAB format [12]. The `extChan` field represents external channels with the same structure to the data field. `cndVersion` field, records the version number of the CND specification. The data within the `dataCND` folder needs to be synchronised, making sure that each time sample in the neural data corresponds to the same time sample in the stimulus data.

The CND structure differs from other data formats (e.g. Brain Imaging Data Structure - BIDS [17]), rather than only general-purpose raw data storage and experimental paradigms, CND focusing on higher-level ready-to-analyse data that aligns directly to specialised tools like the mTRF-Toolbox [10].

Conversion and Modifications for This Project

The process of converting our pre-processed EEG data to the CND format modified for this project’s experimental design. It involves two MATLAB scripts,

`SL_create_CND_dataStim.m` and `SL_create_CND_dataEeg.m`, both of which are modified from the example scripts from CNSP workshop [13, 23].

The `SL_create_CND_dataStim.m` script defines the path of the input and output directories, then sets general parameters, including the numbers of trials per condition (14 trials per subject), the down-sampled frequency (64 Hz), and the conditions under investigation: V and R. The organisation of data is structured into sub-directories within the primary condition folders, reflecting different EEG sub-bands. For each condition, the script generates a separate `dataStim.mat` file, encapsulating the stimulus features for that condition. It iterates through the predefined feature names and corresponding stimulus files. Nan values within the stimulus suggests that trial was not presented to the subject (we randomly play 11 in correct playback and 3 trials in time-reversed), they are set to zero. To align this with analysis pipeline, we have an additional ‘nonemptyTrials’ field introduced to index the trials that contains data.

The `SL_create_CND_dataEeg.m` script handles the conversion of pre-processed EEG data into CND format, it also starts by initialises paths for EEG input and output directories. We have parameters set that includes the number of trials played, and the subjects to select, since the EEG folder contains both LSF and FR cohorts, this parameter is for separate them for comparison analysis.

To address the issues with bad channels, our script includes a step for interpolating these channels using the '`removeBadChannels`' function, this function identifies and removes problematic channels from EEG data.

These two scripts implemented several modifications to the standard CND structure to fit the need of our study, as the diagram 2.3.4 illustrated.

Our CND Data Directory Structure

CND

R

1-30Hz

Mastoids

pre_dataSub1.mat

...

pre_dataSub6.mat

1-4Hz

Mastoids

pre_dataSub1.mat

...

pre_dataSub6.mat

1-8Hz

Mastoids

pre_dataSub1.mat

...

pre_dataSub6.mat

dataStim.mat

V

1-30Hz

Mastoids

pre_dataSub1.mat

...

pre_dataSub6.mat

1-4Hz

Mastoids

pre_dataSub1.mat

...

pre_dataSub6.mat

1-8Hz

Mastoids

pre_dataSub1.mat

...

pre_dataSub6.mat

dataStim.mat

chanlocs64.mat

2.3.5 Ethical Considerations

Our data collection was conducted entirely in France. The procedures and protocols followed strictly adhered to the guidelines and regulations mandated by French and European authorities. The study protocol has been reviewed and approved by the relevant French ethical committee, the “Comité de Protection des Personnes” (CPP). The protocol was designed to ensure the maximal protection of the participants while contributing to the advancement of scientific knowledge without foreseeable risks to the volunteers.

All participants involved in our study have provided informed written consent. Prior to participating, each individual received a comprehensive information sheet of the study’s aims, procedures, potential risks, and benefits. This information was presented both orally and in written form, ensuring clarity and comprehension. Specifically, the consent forms adhered to the methodological reference MR-003 and were compliant with the General Data Protection Regulation (GDPR) and the French “Loi Informatique et Libertés”.

Participant confidentiality was maintained through the use of anonymised data coding systems. Each participant was assigned a unique code, ensuring that personal information could not be traced back to them without access to a secure, confidential log held by the primary investigator. Importantly, the identification key linking participant names to their codes was destroyed by our French collaborator. With the destruction of this key, the data falls outside of GDPR, thereby negating the need for a data sharing agreement between the French institution and Trinity College Dublin.

All collected data were stored securely and managed in accordance with French and European data protection regulations, including the GDPR. Access to data was strictly limited to authorised personnel, and regular quality control measures were implemented to ensure ongoing compliance.

The study was conducted in accordance with the Declaration of Helsinki and Good Clinical Practices (GCP). Our protocol explicitly states that the research holds no direct benefits or foreseeable risks for the participants. This aspect was critically assessed by the ethical committee, who confirmed the ethical soundness and regulatory compliance of our study design.

To ensure ethical standards and to compensate participants for their time and efforts, the collaborators at PSL University are paying every participant €15 per hour, regardless of whether participants complete the entire session or choose to withdraw early. This remuneration is reflective of the time they spend participating in the study, including behavioural and electroencephalography (EEG) testing.

The project has received necessary approvals in France and is pending further approval from Trinity College Dublin (reference number #3559 in the research ethics application

management system (REAMS), currently Awaiting Recommendation as a Risk 2 application). This adherence to a robust ethical framework underscores the integrity and responsibility foundational to our research approach.

Chapter 3

Results & Evaluation

In this chapter, we aim to present the results from different experimental conditions, we first present the results using IVC as feature, then show the result with the new OpenPose-IVC.

3.1 Global Field Power (GFP)

Global Field Power (GFP) is a reference-independent metric used in EEG to quantify the overall strength of the brain's electrical activity across the scalp. It provides a measure of neural response strength by capturing the variance of electric potentials recorded from all 64 electrodes. The calculation of GFP begins with the collection of EEG signals from multiple scalp electrodes over a period of time, in our project, based on previous studies on speech and sound perception [21], we set our time lag to -50 to 550 milliseconds. The time lag refers to the time interval between the presentation of a stimulus and the observation of a neural response.

Let $V_i(t)$ denote the signal from electrode i at time t , where $i = 1, 2, \dots, N$ and N is the total number of electrodes. For each time point t , the mean voltage across all electrodes is calculated as follows:

$$\bar{V}(t) = \frac{1}{N} \sum_{i=1}^N V_i(t) \quad (3.1)$$

Subsequently, the GFP at each time point t is computed by taking the square root of the sum of the standard deviations of each electrode's voltage from the mean voltage:

$$\text{GFP}(t) = \sqrt{\frac{1}{N} \sum_{i=1}^N (V_i(t) - \bar{V}(t))^2} \quad (3.2)$$

As shown in Figure 3.1, the shape of GFP for the signers group (LSF) and French

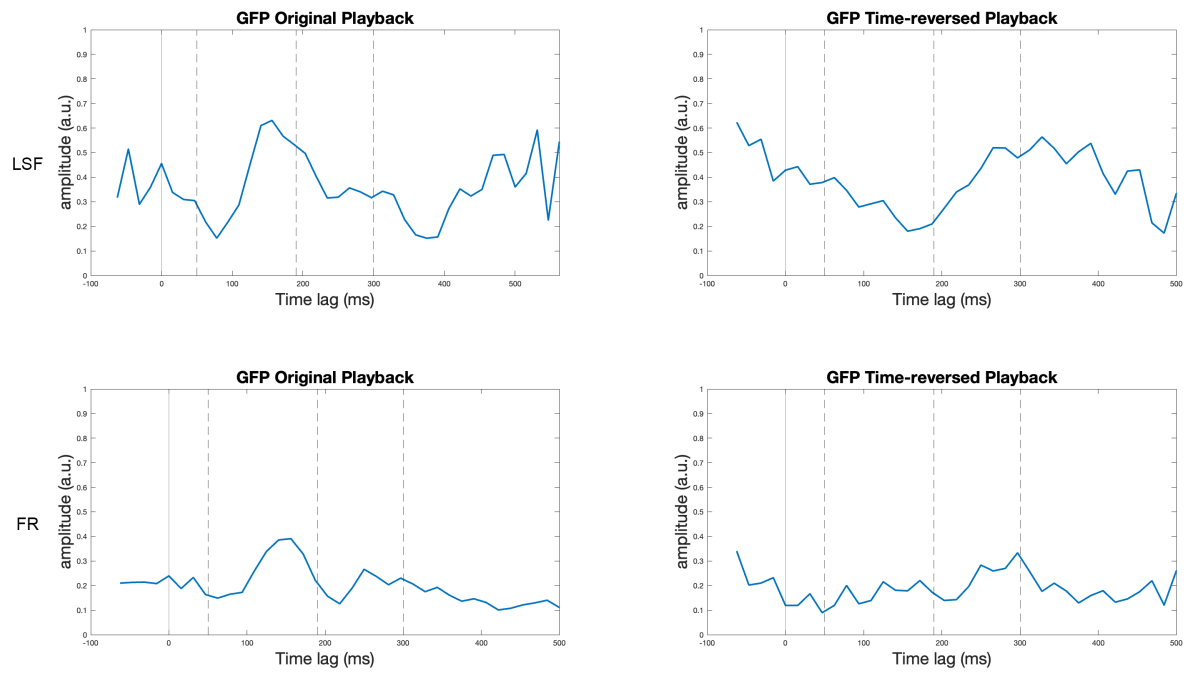


Figure 3.1: Global Field Power (GFP) amplitude over time for French Sign Language signers (LSF) and French monolingual speakers (FR) during original and time-reversed playback conditions at 1-8 Hz EEG sub-band

monolingual speaker group (FR) reveals different patterns in neural response to both original and time-reversed playback conditions.

3.1.1 LSF Cohort

For the LSF group, the GFP amplitude during the original playback condition observes a dynamic pattern over the time lag from -50 to 550 milliseconds. The amplitude initiates at approximately 0.3 arbitrary units (a.u.) and demonstrates several peaks and troughs throughout the time course. Notable peaks are observed around 100 ms - 200 ms, and 500 ms, with the highest peak reaching approximately 0.7 a.u.. In contrast, the time-reversed playback condition for the LSF group shows a different pattern. The GFP amplitude starts higher, around 0.4 a.u., and displays a gradual decline with minor peaks around 100 ms and 300 ms. The highest peak in this condition is around 0.6 a.u. at 320 ms, suggesting a less dynamic response compared to the original playback.

3.1.2 FR Cohort

In the FR group, the GFP amplitude during the original playback condition starts at approximately 0.2 a.u. and maintains a relatively stable pattern with minor fluctuations. A noticeable peak occurs around 150 ms, reaching about 0.4 a.u., followed by a decline and another minor peak around 300 ms. This indicates a less variable neural response compared to the LSF group. During the time-reversed playback condition, the GFP amplitude for the FR group starts at around 0.2 a.u. and remains relatively low and stable throughout the time lag. Minor peaks are observed around 300 ms, but the overall amplitude does not exceed 0.4 a.u., indicating a more subdued neural response to the time-reversed playback compared to the original playback.

3.1.3 Statistical tests for GFP results

The Wilcoxon signed-rank test was selected to compare the GFP values within each cohort (LSF and FR) between the normal playback and time-reversed playback conditions. This non-parametric test is appropriate for paired data and does not assume a normal distribution, making it suitable for our EEG data, which can exhibit non-normal characteristics. By using the signed-rank test, we aimed to determine whether the temporal manipulation of the sign language videos (normal vs. time-reversed playback conditions) elicited different neural responses within the same group of participants.

On the other hand, the Wilcoxon rank-sum test (also known as the Mann-Whitney U test) was chosen to compare the GFP values between the two independent cohorts (LSF and FR) for each playback condition. Similar to the signed-rank test, this is also a non-parametric test and does not require the assumption of normality. The rank-sum test allowed us to assess whether the neural responses to the sign language videos differed significantly between the LSF group and the FR group. By employing these two complementary statistical tests, we were able to comprehensively analyse the GFP data, addressing both within-group and between-group differences in neural responses.

Wilcoxon Signed-Rank Test

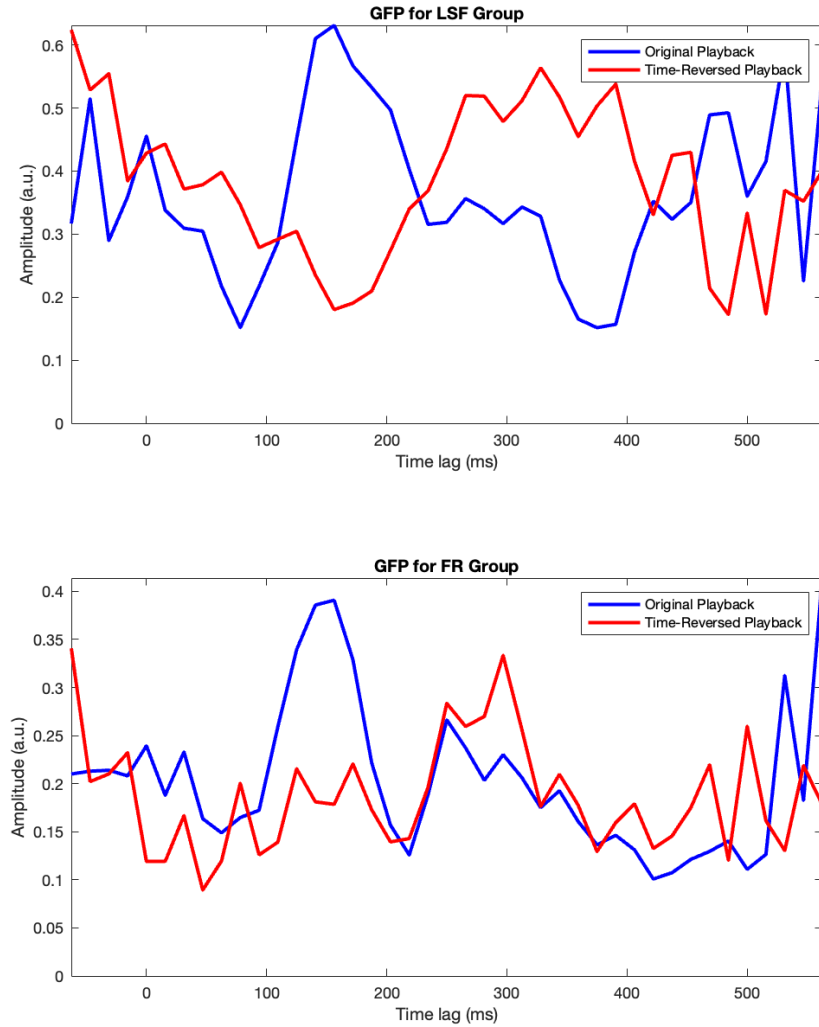


Figure 3.2: Wilcoxon signed-rank test for GFP results at 1-8 Hz EEG sub-band

Table 3.1: Results of the Wilcoxon Signed Rank Test

Group	Condition	p -value
LSF	Normal Playback (V) vs. Time-reversed Playback (R)	0.3817
FR	Normal Playback (V) vs. Time-reversed Playback (R)	0.6362

The Wilcoxon signed-rank test was conducted to compare the GFP values within each cohort (LSF and FR) between the normal playback (V) and time-reversed playback (R) conditions. For the LSF group, the test yielded a p -value of 0.3817 ($p > 0.05$) and a test statistic of 363.0000, indicating no significant difference in GFP values between the normal and time-reversed playback conditions. Similarly, for the FR group, the test produced a p -value of 0.6362 ($p > 0.05$) and a test statistic of 467.0000, also suggesting no significant difference in GFP values between the two playback conditions.

Wilcoxon Rank-Sum Test

Table 3.2: Results of the Wilcoxon Rank Sum Test on GFP Values

Group	Condition	p -value	statistic
LSF vs. FR (V)	Normal Playback (V)	0.0000	1109.0000
LSF vs. FR (R)	Time-reversed Playback (R)	0.0000	992.0000

The Wilcoxon rank-sum test was employed to compare the GFP values between the LSF and FR groups for both normal playback (V) and time-reversed playback (R) conditions. For the normal playback condition, the test revealed a highly significant difference between the two groups, with a p -value of 0.0000 ($p < 0.05$) and a test statistic of 1109.0000. Similarly, for the time-reversed playback condition, the test also indicated a highly significant difference, with a p -value of 0.0000 ($p < 0.05$) and a test statistic of 992.0000. These results suggest that the neural responses, as measured by GFP, differ significantly between the LSF and FR groups under both playback conditions.

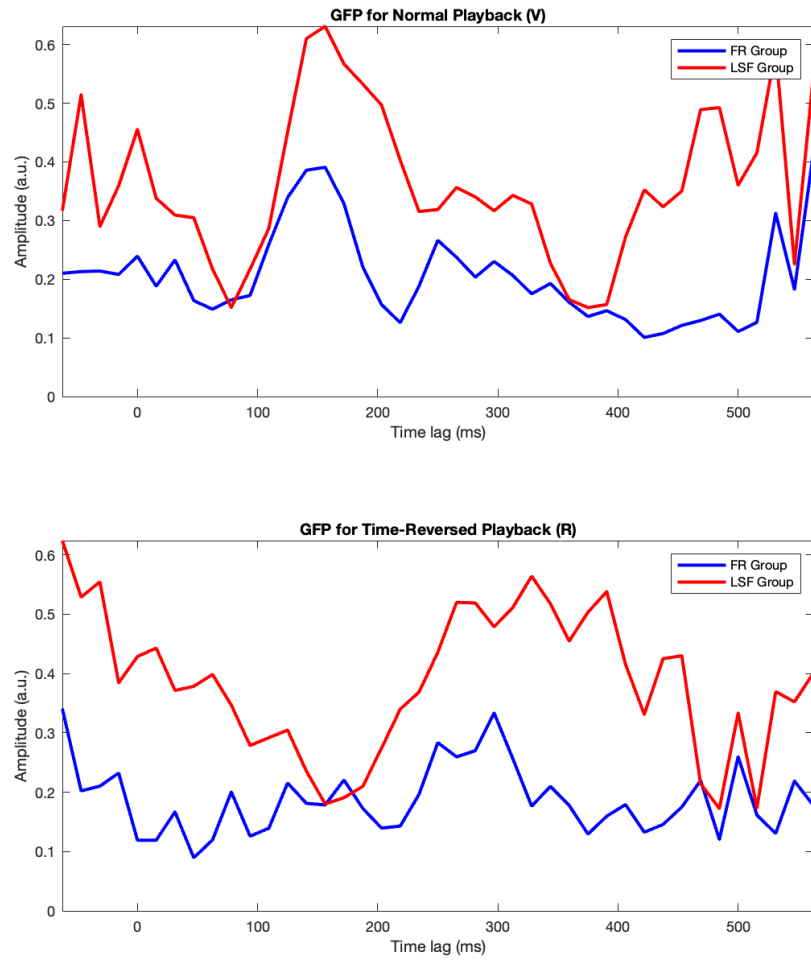


Figure 3.3: Wilcoxon signed-rank test for GFP results at 1-8 Hz EEG sub-band

3.2 TRF

3.2.1 TRF regression weights

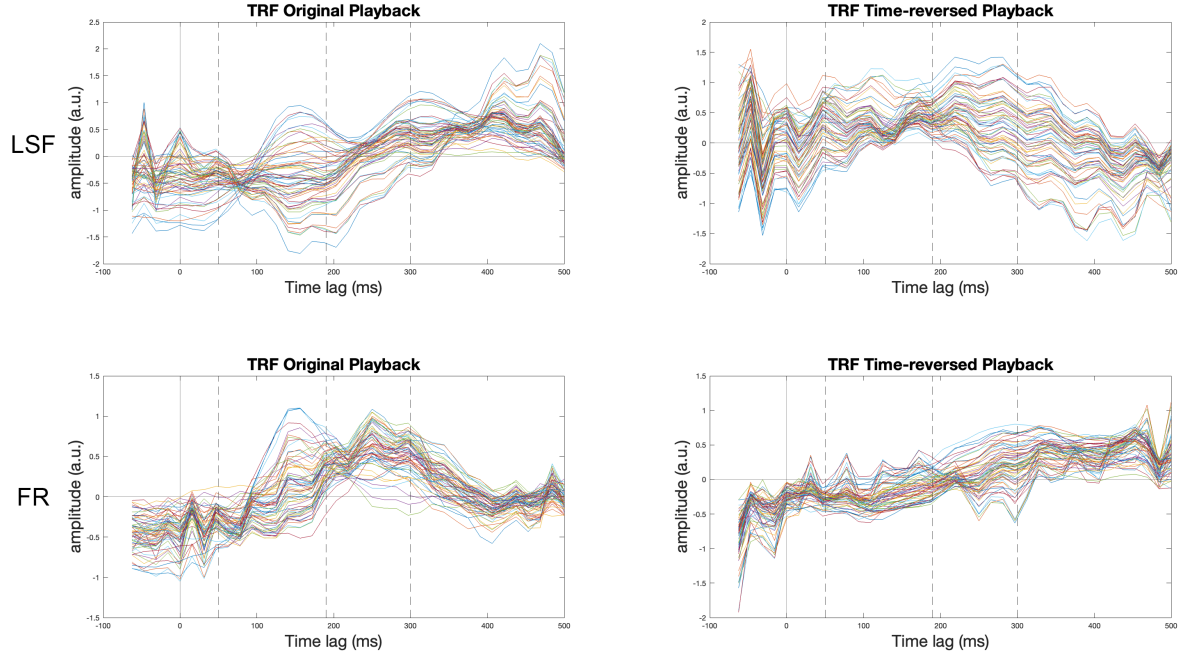


Figure 3.4: Temporal Response Functions (TRFs) for LSF and FR groups under original playback and time-reversed playback conditions.

The TRFs are averaged across participants within each cohort. TRFs are trained with EEG sub-band 1-8 Hz.

The Temporal Response Functions (TRFs) for both the French Sign Language (LSF) signers and the French speaker (FR) groups were analysed under two conditions: original playback (V) and time-reversed (R) playback of sign language video stimuli. The TRFs were averaged across participants within each cohort to obtain representative response profiles. Each line in the graphs represents the TRF regression weights from a different electrode, capturing the neural response at various scalp locations.

LSF Cohort

In the LSF cohort, the TRFs for the original playback condition (Figure 3.4, top left) exhibited a distinct pattern of neural activity. The amplitude of the TRFs showed a prominent peak around 200 ms, followed by a gradual decline. This peak suggests a robust neural encoding of the sign language stimuli in the original playback condition

(V). In contrast, the TRFs for the time-reversed playback condition (Figure 3.4, top right) displayed a more attenuated response, with less pronounced peaks and a generally lower amplitude across the time lags. This indicates a reduced neural response to the temporally altered stimuli.

FR Cohort

For the FR cohort, the TRFs in the original playback condition (Figure 3.4, bottom left) also demonstrated a notable dual peak around 200 ms, similar to the LSF cohort, albeit with a slightly lower amplitude. This might suggest that even non-signers exhibit some degree of neural response to sign language stimuli, potentially due to the ability of human brain to predict body motion and rhythmic patterns inherent in sign language. However, the TRFs for the time-reversed playback condition (Figure 3.4, bottom right) in the FR cohort showed a marked reduction in amplitude and a less defined peak, mirroring the pattern observed in the LSF cohort.

3.2.2 Topographical plots

The topographical plots of the TRF weights for both the LSF and FR groups were grouped under two conditions: original playback and time-reversed playback of sign language video stimuli, shown in Figure 3.5. These plots provide a spatial representation of the neural response across the scalp at various time lags, offering insights into the temporal dynamics and spatial distribution of neural processing in response to the stimuli.

LSF Cohort

In the LSF cohort, the topographical plots for the original playback condition (top two row) reveal a dynamic pattern of neural activity across the scalp. At early time lags (e.g., -50 ms to 0 ms), there is a notable activation in the frontal regions, which gradually diminished. At 300 ms, a strong positive activation is observed in the rear regions, which peaks around 350 ms and then gradually shifts to central and frontal region. This pattern suggests a robust and temporally coherent neural encoding of the sign language stimuli, with the most significant activity occurring in the rear and central regions, likely reflecting the processing of visual motion and linguistic information inherent in sign language.

In contrast, the time-reversed playback condition (Figure 3.5, row 3 and 4) shows a markedly different pattern. The neural responses are more pronounced and more diffuse across the frontal and central regions from -50 to 150ms, then the shift towards rear region begins earlier at around 200 ms time lag.

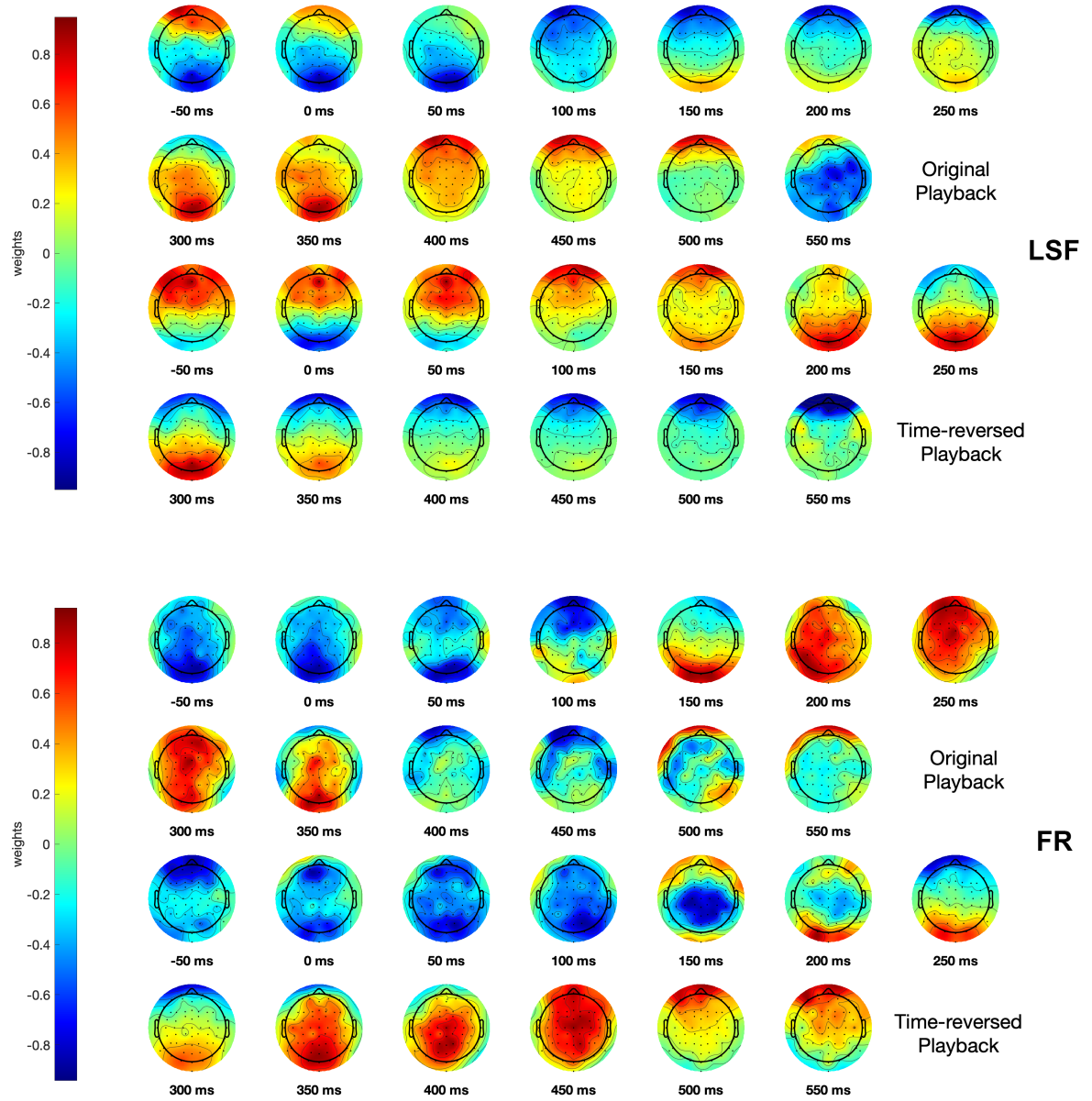


Figure 3.5: Topographical plot of TRF weights (1-8 Hz) across cohorts and conditions with 50 milliseconds interval

Top half: TRFs for LSF signers under original playback (V) (top two rows) and time-reversed playback (R) (row 3 and 4). Bottom half: Topographical plots of TRF weights for FR cohort under original playback (V) (row 5 and 6) and time-reversed playback (R) (row 7 and 8). The plots show the spatial distribution of neural responses across the scalp at various time lags, highlighting the regions of the brain involved in processing the stimuli.

FR Cohort

For the FR cohort, the topographical plots for the original playback condition (Figure 3.5, row 5 and 6) also demonstrate a distinct pattern of neural activity, with some significant differences compared to the LSF cohort. At early time lags, there is a weaker activation in across the scalp compared to both conditions of LSF cohort. The strong rear region activation then shifting to the whole scalp can be observed from 150 ms - 300 ms. This peak at rear region is slightly lower in amplitude compared to the LSF cohort, suggesting that while non-signers do exhibit neural responses to sign language stimuli, these responses are less robust. The activation then gradually diminishes, and some activation on the frontal region can be observed similar to the pattern in the LSF cohort but weaker.

The time-reversed playback condition for the FR cohort (bottom two rows) shows a reduction in neural response amplitude and a less defined spatial distribution. The activation is more scattered at 300 ms to 400 ms, with some rear region activation before this.

Comparative Analysis

Overall, the topographical plot of TRF weights on LSF cohort shows a pattern that first activation at frontal region, then the activation shifts to rear region where the visual cortex is located. The LSF-V condition shows an additional shift towards frontal region after the visual cortex response suggest potentially linguistic processing at subjects that are fluent in sign language. The FR group shows a different, more scattered activation pattern and weaker activation compared to sign language group, suggest that the FR group might have lower attention level during the present of visual language they are not familiar with.

3.3 Prediction correlation

3.3.1 Prediction Correlation Analysis

The prediction correlation graph (Figure 3.6) illustrates the prediction correlations (r) from TRF regression training for different conditions: FR-V, FR-R, LSF-V, and LSF-R. The box plots represent the distribution of prediction correlations for each condition.

FR-V (French speaking cohort with correct playback stimuli) Condition: The prediction correlations for the FR-V condition show a median value slightly above zero, with a range extending from approximately -0.005 to 0.015. The interquartile range (IQR) is relatively narrow, indicating moderate variability in the prediction correlations.

FR-R Condition: The prediction correlations for the FR-R condition have a median value

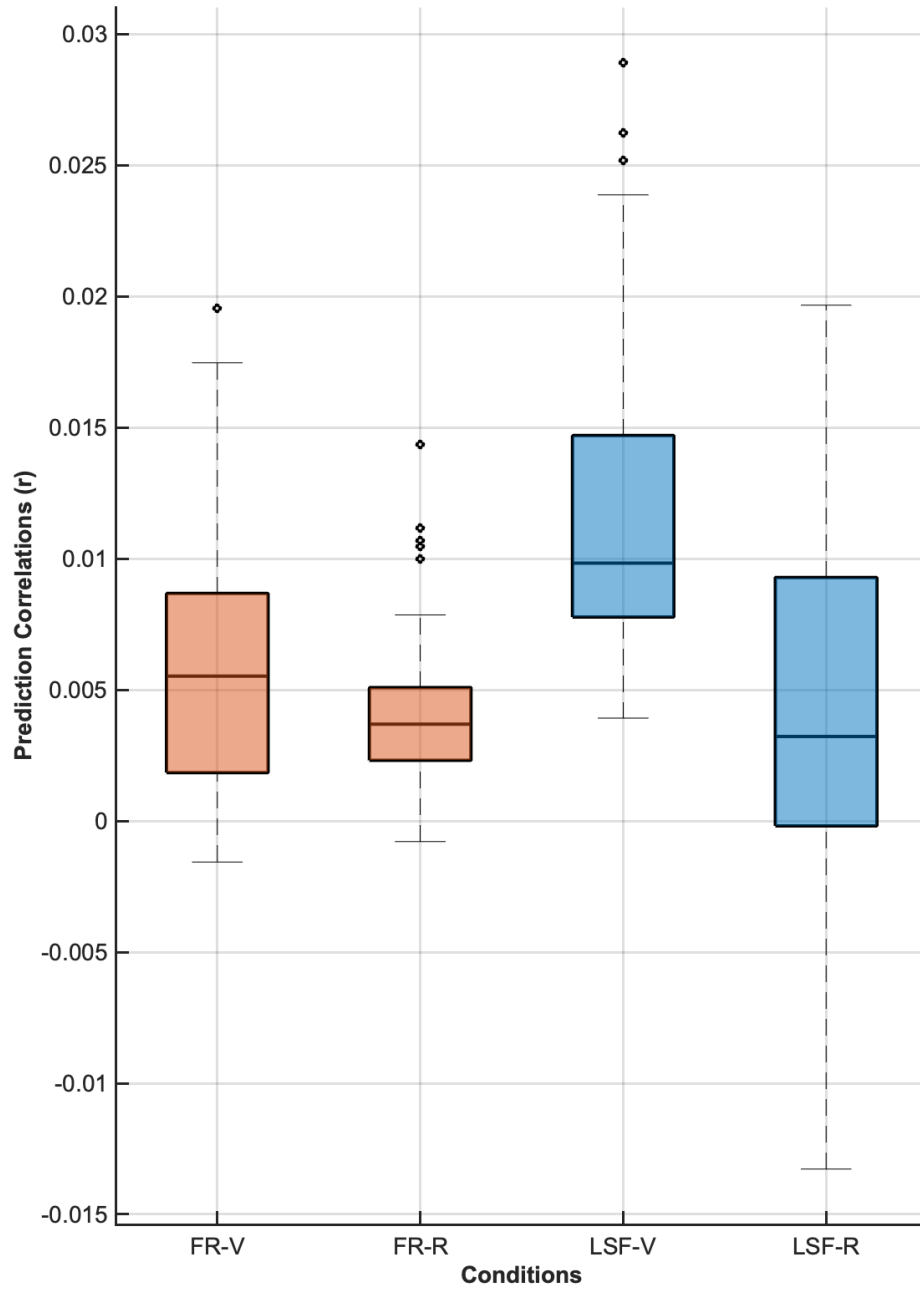


Figure 3.6: Prediction correlations (r) for different conditions (FR-V, FR-R, LSF-V, and LSF-R).

The box plots illustrate the distribution of prediction correlations for each condition, showing higher prediction correlations for the LSF conditions compared to the FR conditions.

close to zero, with a range extending from approximately -0.01 to 0.01. The IQR is narrower compared to the FR-V condition, suggesting lower variability in the prediction correlations.

LSF-V Condition (French signer cohort with correct playback stimuli videos): The prediction correlations for the LSF-V condition exhibit a higher median value compared all three other conditions, with a range extending from approximately 0 to 0.025. The IQR is wider, indicating higher variability in the prediction correlations.

LSF-R Condition: The prediction correlations for the LSF-R condition have a median value slightly above zero, with a range extending from approximately -0.01 to 0.02. The IQR is wider compared to the FR-R condition, suggesting higher variability in the prediction correlations.

The box plots indicate that the LSF conditions (both V and R) generally have higher prediction correlations compared to the FR conditions. This observation is supported by the ANOVA results, which indicated a significant main effect of Condition and a significant interaction effect between Group and Condition. The higher prediction correlations in the LSF conditions suggest that the LSF group may have a stronger or more consistent response to the conditions compared to the FR group.

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob > F
Group	0.00002	1	0.00002	0.73	0.3952
Condition	0.00178	1	0.00178	55.18	0.0000
Group:Condition	0.00019	1	0.00019	5.96	0.0154
Error	0.00813	252	0.00003		
Total	0.01012	255			

Table 3.3: Results of the two-way ANOVA with interaction.

The table shows the sources of variation (Group (FR vs. LSF), Condition (V vs. R),

Group:Condition interaction, Error, and Total), sum of squares, degrees of freedom, mean squares, F-statistics, and associated p-values. The analysis indicates no significant main effect of Group, a significant main effect of Condition, and a significant interaction effect between Group and Condition.

3.3.2 ANOVA Analysis

The results of the two-way ANOVA with interaction are presented in Table 3.3. The analysis was conducted to examine the effects of Group (FR vs. LSF) and Condition (V vs. R) on the response variable. The ANOVA table includes the sources of variation, sum of squares, degrees of freedom, mean squares, F-statistics, and associated p-values.

Group Effect: The sum of squares for the Group effect was 0.00002 with 1 degree of freedom, resulting in a mean square of 0.00002. The F-statistic for the Group effect

was 0.73, with a p-value of 0.3952 ($p > 0.05$). This p-value is greater than the common significance level of 0.05, indicating that there is no statistically significant effect of Group (FR vs. LSF) on the response variable.

Condition Effect: The sum of squares for the Condition effect was 0.00178 with 1 degree of freedom, resulting in a mean square of 0.00178. The F-statistic for the Condition effect was 55.18, with a p-value of 0 ($p < 0.05$). This p-value is less than the common significance level of 0.05, indicating a statistically significant effect of Condition (V vs. R) on the response variable.

Group:Condition Interaction: The sum of squares for the interaction between Group and Condition was 0.00019 with 1 degree of freedom, resulting in a mean square of 0.00019. The F-statistic for the interaction effect was 5.96, with a p-value of 0.0154 ($p < 0.05$). This p-value is less than the common significance level of 0.05, indicating a statistically significant interaction effect between Group and Condition on the response variable. This suggests that the effect of Condition on the response variable depends on the Group.

The ANOVA results indicate that there is no significant main effect of Group on the response variable, but there is a significant main effect of Condition. Additionally, there is a significant interaction effect between Group and Condition, suggesting that the effect of Condition on the response variable is dependent on the Group.

3.4 Discussion

The results of the prediction correlation and ANOVA testing provide a comprehensive understanding of the effects of Group and Condition on the response variable. The significant main effect of Condition indicates that the response variable is influenced by whether the condition is V or R. This suggests that visual tracking is increased when participants are understanding the sign language, which is only one out of the four conditions.

The significant interaction effect between Group and Condition further reveals the relationship between these factors. Specifically, the interaction effect indicates that the relationship between Condition (V vs. R) and the response variable changes across the two groups (FR vs. LSF). In practical terms, this means that the effect of Condition on the response variable is different for the FR group compared to the LSF group. This aligns with our hypothesis that the neural response in the signer's group will be the highest and most correlated to the features extracted from the stimuli videos when watching the correctly played back videos. We assume that this group, due to their familiarity with sign language, will exhibit stronger and more specific neural entrainment to the temporal and linguistic properties of the sign language stimuli compared to the non-signers' group. The prediction correlation analysis supports these findings by showing higher prediction

correlations for the LSF conditions compared to the FR conditions. This suggests that the LSF group has a stronger or more consistent response to the conditions, which is consistent with the significant interaction effect observed in the ANOVA analysis.

The higher variability observed in the LSF conditions could be attributed to the smaller sample size of subjects compared to the FR conditions. Specifically, there are 16 FR subjects and only 6 LSF subjects. This smaller sample size in the LSF group may lead to greater variability in the prediction correlations.

Additionally, the slightly higher prediction correlation in the FR-V condition compared to the FR-R condition might be due to the human brain’s ability to predict body motions. Research suggests that the human brain is adept at anticipating and predicting the movements of others, which could explain the higher prediction correlations when participants are viewing correctly played back videos (FR-V) as opposed to time-reversed videos (FR-R). This ability to predict body motions is supported by studies on action observation and motor prediction, which indicate that the brain’s motor system is engaged when observing actions, facilitating the prediction of subsequent movements [20].

In conclusion, the results indicate that there is a significant effect of Condition on the response variable and that this effect is modulated by Group. The significant interaction effect suggests that the relationship between Condition and the response variable is different for the FR and LSF groups, which is consistent with our hypothesis that visual tracking is increased only when participants are understanding the sign language. The prediction correlation analysis further supports these findings, highlighting the differences in response between the FR and LSF groups under different conditions.

3.5 Performance of New Metrics

3.5.1 TRF Regression Weights

LSF Cohort

For the LSF cohort, the TRFs using the OpenPose-IVC metric in the original playback condition exhibited erratic and inconsistent patterns compared to the IVC metric results. While the IVC metric showed a distinct peak around 200 ms, indicating robust neural encoding, the OpenPose-IVC TRFs lacked this clear peak and displayed high variability. This discrepancy, according to Crosse et al. [11], suggests that the OpenPose-IVC feature extraction may suffer from temporal misalignment, where the feature timing might not accurately correspond to the stimulus timing, leading to a less reliable neural signal, another cause could be lack of training data, this reason was ruled out with the comparatively better results using IVC as feature extraction metric. Additionally, in the time-reversed

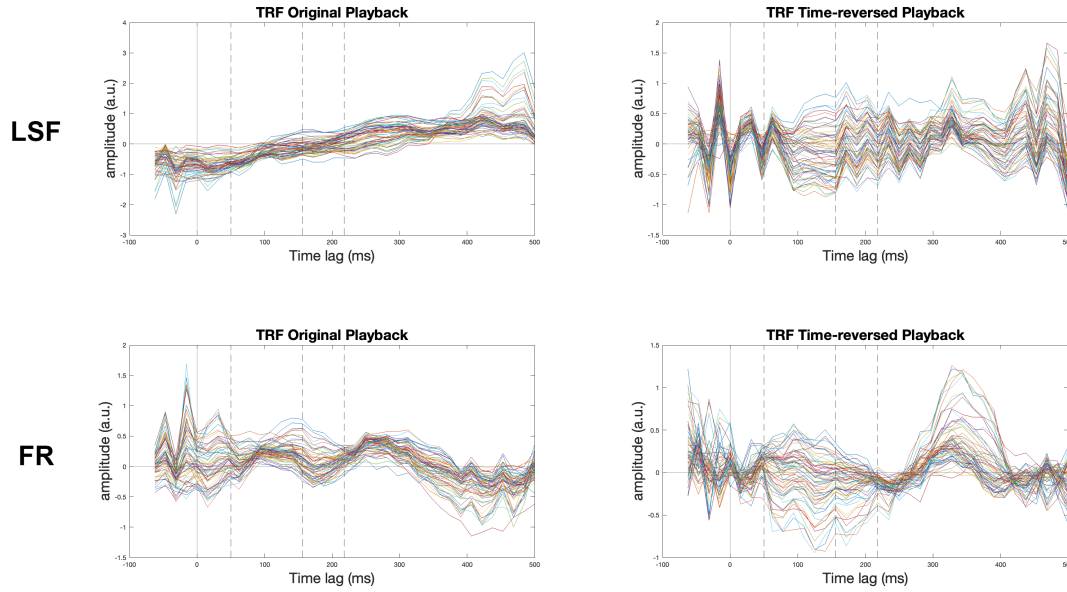


Figure 3.7: Temporal Response Functions (TRFs) for LSF and FR groups using OpenPose-IVC

condition, the OpenPose-IVC TRFs appeared even more attenuated and noisy. In stark contrast to the IVC metric, which retained identifiable response patterns albeit reduced in amplitude, the OpenPose-IVC results suggest a significant reduction in neural response coherence.

FR Cohort

In the FR cohort, the TRFs generated using the OpenPose-IVC metric during the original playback condition also showed significant inconsistencies. Unlike the IVC metric, which produced a notable dual peak around 200 ms, the OpenPose-IVC TRFs were characterised by erratic fluctuations and lacked coherent structure. The time-reversed playback condition exhibited even greater noise and diminished coherence with the OpenPose-IVC metric. The IVC metric, in comparison, demonstrated some definable peaks even with lower amplitude, showing a more structured neural response despite the temporal reversal of stimuli.

Comparing the two metrics, the IVC results displayed better-defined peaks and coherent patterns of neural response in both cohorts and playback conditions. In contrast, the OpenPose-IVC metric led to TRFs with high variability, little defined peaks, and greater noise, indicating several potential issues.

3.5.2 Topographical Plots

LSF Cohort

For the LSF cohort, the OpenPose-IVC topographies under the original playback condition exhibit regions of increased activation, similar to those observed in the IVC plots. However, the patterns in OpenPose-IVC are more sporadic and less concentrated. The characteristic peaks around 150 ms and 200 ms, which were prominent in the IVC results, appear less defined in OpenPose-IVC. When examining the time-reversed playback condition, the activation patterns in OpenPose-IVC are markedly more diffuse and inconsistent, with overall lower amplitude.

FR Cohort using OpenPose-IVC

For the FR cohort, the OpenPose-IVC plots similarly show significant inconsistencies. The expected activation peaks around 150 ms and 200 ms are not as pronounced or focused as those seen in the IVC metrics. Under the time-reversed playback condition, the neural activity captured by OpenPose-IVC appears a lack of distinct high-activation areas, in contrast to the more organised patterns exhibited by the IVC metric.

3.5.3 Discussion

The TRFs using the OpenPose-IVC metric exhibited high variability and inconsistent patterns across both LSF and FR cohorts. Unlike the IVC metric, which showed distinct peaks around 150 ms and 200 ms indicating robust neural encoding, the OpenPose-IVC TRFs lacked such clear peaks. This inconsistency likely suggest issues in the feature extraction process. OpenPose-IVC aims to leverage pose estimation to capture body movements to sign language, potentially offering a more refined approach than the global differences captured by IVC. However, it seems that while OpenPose-IVC can reduce noise by focusing on relevant movements, it might simultaneously miss finer details critical to the nuances of sign language, such as intricate hand shapes and facial expressions. This lack of detail could lead to incomplete feature representation, producing erratic and less reliable neural response patterns. Due to the lower frame rate (30 frames per second) of the stimuli, when there's fast movement, it shows blurry in the frame that caused inaccuracy in pose estimation, this would result in large differences when calculate IVC.

The time-reversed playback condition further highlighted these issues. For both the LSF and FR cohorts, the OpenPose-IVC TRFs in the reversed condition were significantly more attenuated and noisier compared to those in the original condition, and also in comparison to the IVC metric which still retained some identifiable neural response patterns.

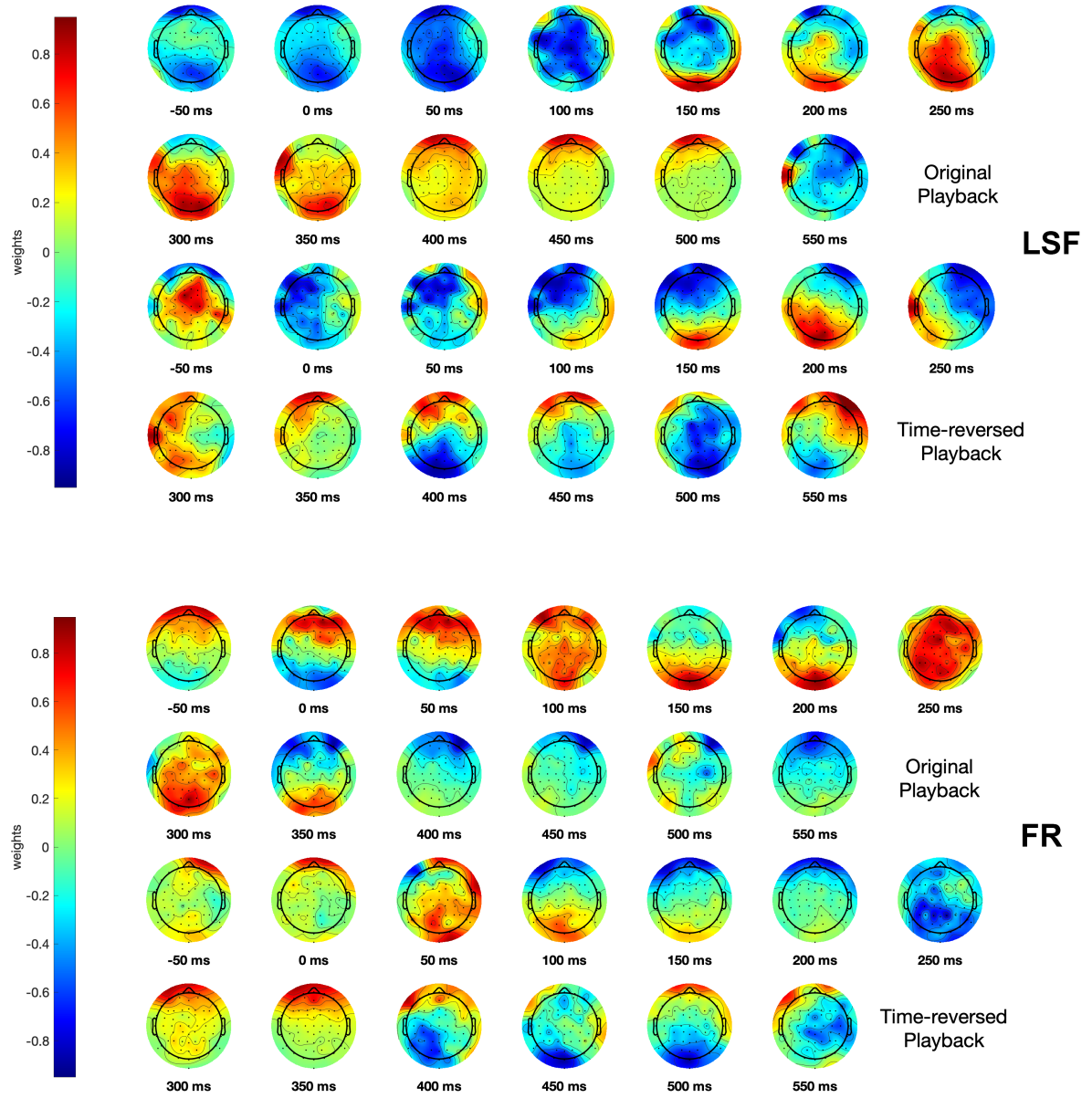


Figure 3.8: Topographical plot of TRF weights (1-8 Hz) across cohorts and conditions with 50 milliseconds interval using OpenPose-IVC feature

This suggests that the OpenPose-IVC features may suffer from temporal misalignment, where the timing of feature data does not accurately match the stimulus timing, further disrupting the neural encoding process.

Analysing the topographical plots, under the original playback condition, the OpenPose-IVC topographies for both cohorts displayed regions of activation similar to those in the IVC plots, but these regions were more sporadic and less concentrated. The characteristic peaks seen in the IVC metric at around 150 ms and 200 ms were less defined in the OpenPose-IVC plots, again pointing to a weaker and more inconsistent neural encoding of the stimuli.

Despite these challenges, the targeted approach of OpenPose-IVC remains promising. By fine-tuning the feature extraction process to better capture the subtleties of sign language movements – integrating higher resolution and accuracy, and employing robust pre-processing to align temporal features accurately – it might be possible to realise the full potential of OpenPose-IVC. Such refinements could lead to cleaner, more precise feature representations and ultimately more reliable and coherent TRF analyses, potentially surpassing the performance of the IVC metric by providing richer neural response data.

Chapter 4

Conclusions & Future Work

This study aimed to investigate the neural responses to continuous sign language stimuli using EEG data. We hypothesised that the neural response in the LSF signers' group would be the highest and most correlated to the features extracted from the stimuli videos when watching correctly played back videos. Our approach involved comparing EEG responses between a group of LSF signers and a group of French monolingual speakers (FR), under both normal and time-reversed playback conditions.

The Global Field Power (GFP) analysis revealed that the LSF cohort exhibited dynamic GFP amplitude patterns during the original playback, with several peaks observed. In contrast, the time-reversed playback condition showed less dynamic GFP patterns in the LSF group, with a gradual decline in amplitude. For the FR group, the GFP amplitude remained relatively stable with minor fluctuations in both playback conditions, indicating less variable neural responses compared to the LSF group.

The Temporal Response Functions (TRF) analysis demonstrated distinct patterns of neural activity for both the LSF and FR groups in the original playback condition, with the LSF group displaying robust peaks around 200 ms. Conversely, the time-reversed condition resulted in more attenuated TRF responses for both groups. These results suggest that even non-signers exhibit some degree of neural response to sign language stimuli, potentially due to the human brain's ability to predict body motion and rhythmic patterns inherent in sign language.

Prediction correlation analysis revealed higher prediction correlations for the LSF conditions compared to the FR conditions, which supports the hypothesis that the LSF group has a stronger response to the sign language stimuli. Furthermore, ANOVA results showed a significant main effect of the Condition and a significant interaction effect between Group and Condition, indicating that the effect of Condition on the response variable depends on the Group.

The evaluation of new metrics, the OpenPose-IVC feature, shows inconsistencies and higher variability compared to the Instantaneous Visual Change (IVC) metric. Despite these challenges, the OpenPose-IVC approach remains promising and merits further refinement to improve performance in capturing sign language movements.

The findings of this study demonstrate that the LSF group, familiar with sign language, exhibits stronger and more specific neural entrainment to the temporal and linguistic properties of the sign language stimuli compared to the FR group. This supports our hypothesis that the brain processes these linguistic signals distinctly when there is familiarity with the language.

Future research could focus on refining the OpenPose-IVC feature extraction process to better align temporal features and capture the nuances of sign language more accurately, which was out of the scope of this project due to the time limit. Furthermore, exploring more robust pre-processing techniques and increasing the sample size could lead to more reliable and generalisable results. The ongoing data collection, which will also include word-by-word transcription of the sign language stimuli videos, holds promise for providing a richer dataset. This transcription will serve as an additional linguistic feature on top of the existing visual ones, potentially revealing deeper insights into neural processing. With the complete dataset, we can expect to obtain better results and uncover deeper findings regarding the neural mechanisms of sign language comprehension. Specifically, the addition of detailed linguistic features could elucidate the interplay between visual and linguistic processing in the brain, offering a more comprehensive understanding of how sign language is perceived and processed by signers and non-signers alike.

This study contributes to the understanding of the neural mechanisms of sign language processing and highlights the importance of considering the temporal and linguistic properties of stimuli in neural entrainment studies.

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Appendix 1 - GenAI Usage

I hereby declare that I have utilised Generative AI (GenAI) tools in the preparation of this dissertation. Specifically, I employed various Large Language Models (LLMs) to gain a deeper understanding of background concepts in neural science, a field outside of my primary academic background in computer science. Additionally, I used GenAI to refine the code for graphs and to assist with proofreading my writing for clarity and coherence. However, I affirm that the intellectual work, including the conception, design, and execution of the research, is entirely my own. The GenAI tools were used solely to augment my knowledge and to improve the presentation of my work, and not to replace or diminish my own intellectual contributions.

I acknowledge the capabilities and limitations of GenAI and take full responsibility for the content and accuracy of this dissertation.

Appendix 2 - Data Protection Training Certificate



Figure 4.1: Data Protection Training Certificate